Adjoint-DSMC Parameter Study

Hansel Carmona

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1 Run Settings

Table 1: Every optimisation run. The drag reductions are relative to the circle in percent. Own is measured on the run's own seeds, and Unseen is re-measured on 16 held-out seeds at 250k particles. The pass-4 run took 10 endgame steps after its 30 normal iterations, and its own-seed value is relative to its starting shape rather than the circle.

Pass	Run	N_REF	N_AVG	N_FOURIER	GRAD_INIT _FRAC	N_ITER	SEED0	Iterations done	Stopped at	Own %	Unseen %	Hours
1	B0_seed0	250k	4	10	0.05	20	0	20	it. limit	51.9	51.8	0.6
1	nseeds1	250k	1	10	0.05	20	0	20	it. limit	50.8	51.0	0.5
1	nseeds16	250k	16	10	0.05	20	0	20	it. limit	51.4	51.5	0.8
1	nseeds2	250k	2	10	0.05	20	0	20	it. limit	52.1	52.0	0.6
1	nseeds8	250k	8	10	0.05	20	0	20	it. limit	51.9	51.7	0.6
1	nref100k	100k	4	10	0.05	20	0	20	it. limit	49.7	50.2	0.2
1	nref1M	1M	4	10	0.05	20	0	20	it. limit	51.1	51.1	1.8
1	nref500k	500k	4	10	0.05	20	0	20	it. limit	52.6	52.8	0.9
1	nf20	250k	4	20	0.05	20	0	3	noise limit	10.2	–	0.2
1	nf3	250k	4	3	0.05	20	0	11	noise limit	32.4	32.6	0.3
1	nf5	250k	4	5	0.05	20	0	20	it. limit	49.2	49.0	0.4
1	frac0.01	250k	4	10	0.01	20	0	20	it. limit	51.2	51.5	0.6
1	frac0.02	250k	4	10	0.02	20	0	20	it. limit	52.6	52.4	0.6
1	frac0.1	250k	4	10	0.1	20	0	20	it. limit	52.4	52.6	0.7
1	frac0.2	250k	4	10	0.2	20	0	20	it. limit	53.8	53.5	0.6
2	B0_seed0	250k	4	10	0.05	50	0	50	it. limit	61.8	62.0	1.6
2	B0_seed100	250k	4	10	0.05	50	100	32	noise limit	57.9	58.3	1.1
2	B0_seed200	250k	4	10	0.05	50	200	27	noise limit	55.0	55.4	0.9
2	nseeds1_seed0	250k	1	10	0.05	50	0	31	noise limit	54.4	53.7	0.9
2	nseeds1_seed100	250k	1	10	0.05	50	100	35	noise limit	50.5	51.2	1.0
2	nseeds1_seed200	250k	1	10	0.05	50	200	24	noise limit	51.4	50.4	0.8
2	nseeds8_seed0	250k	8	10	0.05	50	0	47	noise limit	61.2	61.4	1.5
2	nseeds8_seed100	250k	8	10	0.05	50	100	50	it. limit	62.9	62.9	1.5
2	nseeds8_seed200	250k	8	10	0.05	50	200	50	it. limit	61.7	62.2	1.7
2	nref500k_seed0	500k	4	10	0.05	50	0	50	it. limit	62.3	62.7	2.7
2	nref500k_seed100	500k	4	10	0.05	50	100	50	it. limit	62.5	62.7	2.8
2	nref500k_seed200	500k	4	10	0.05	50	200	50	it. limit	61.8	62.3	2.8
2	nf5_seed0	250k	4	5	0.05	50	0	22	noise limit	49.7	49.4	0.6
2	nf5_seed100	250k	4	5	0.05	50	100	20	noise limit	48.8	49.5	0.4
2	nf5_seed200	250k	4	5	0.05	50	200	22	noise limit	48.9	49.6	0.4
2	frac0.1_seed0	250k	4	10	0.1	50	0	42	noise limit	60.2	60.1	1.3
2	frac0.1_seed100	250k	4	10	0.1	50	100	39	noise limit	60.7	60.4	1.2
2	frac0.1_seed200	250k	4	10	0.1	50	200	39	noise limit	59.7	59.6	1.2
3	grow4to32_seed0	250k	4→32	10	0.05	100	0	81	noise limit	68.5	68.1	4.1
3	grow4to32_seed100	250k	4→32	10	0.05	100	100	100	it. limit	68.7	69.0	11.6
3	grow4to32_seed200	250k	4→32	10	0.05	100	200	88	noise limit	68.4	68.6	6.1
3	grow4to32_seed300	250k	4→32	10	0.05	100	300	100	it. limit	68.7	68.5	8.3
4	C3_1M_endgame_seed400	1M	16→32	10	0.01	30+10	400	20	noise limit	0.4	68.9	6.3

2 Summary figures

2.1 Pass 1

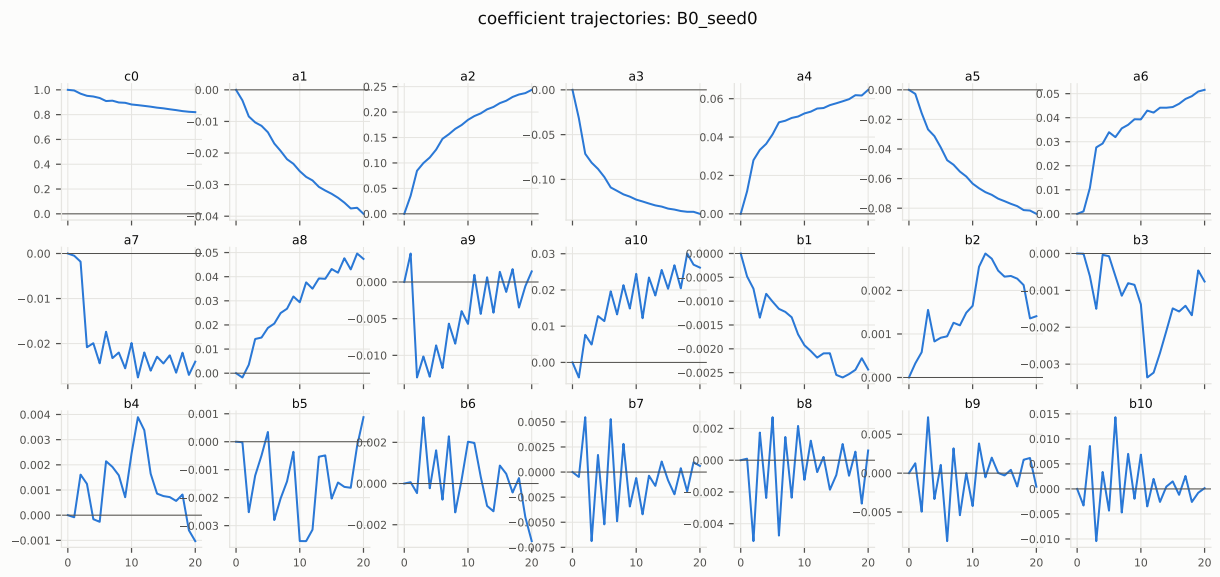


Figure 1: This figure shows each Fourier coefficient of the baseline run with SEED0 = 0 against iteration. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

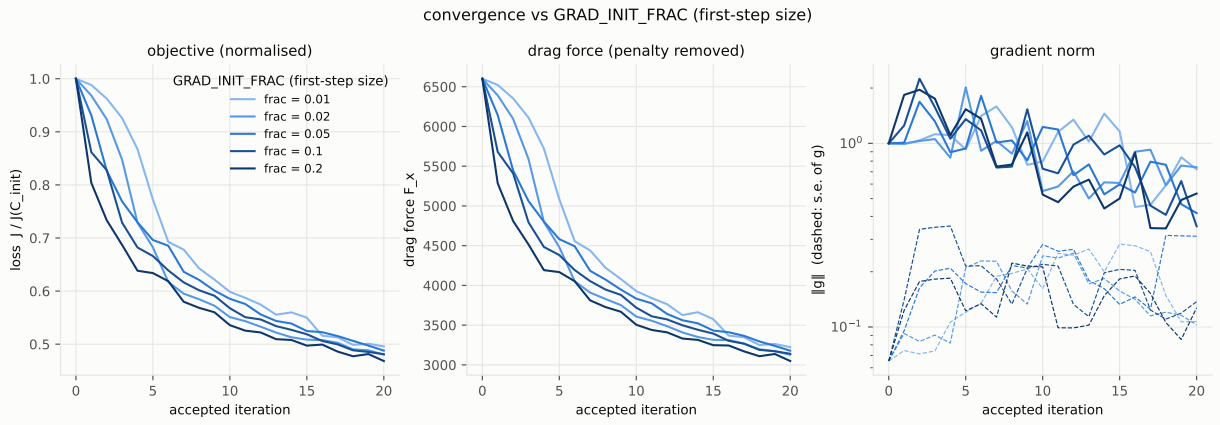


Figure 2: This figure shows the objective and the gradient norm against iteration. The runs differ only in GRAD_INIT_FRAC. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

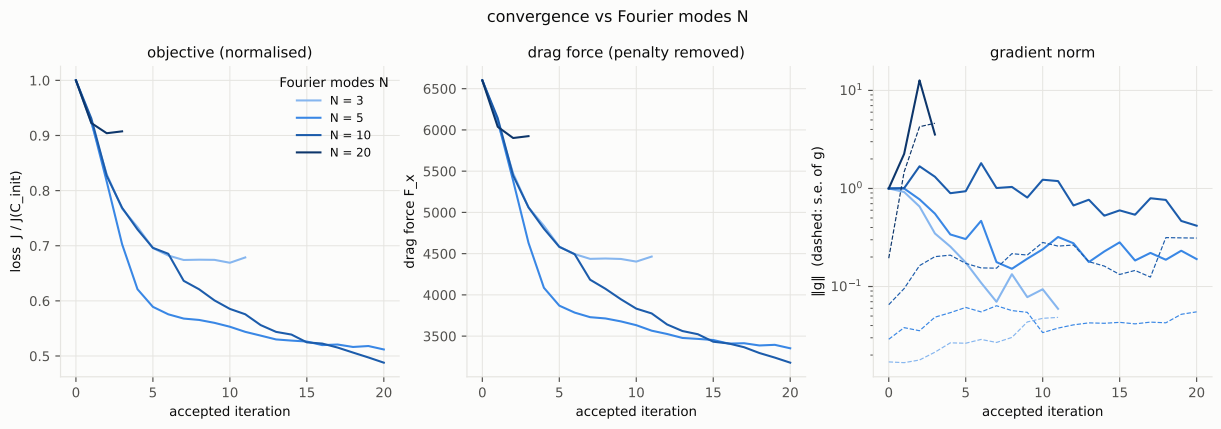


Figure 3: This figure shows the objective and the gradient norm against iteration. The runs differ only in N_FOURIER. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

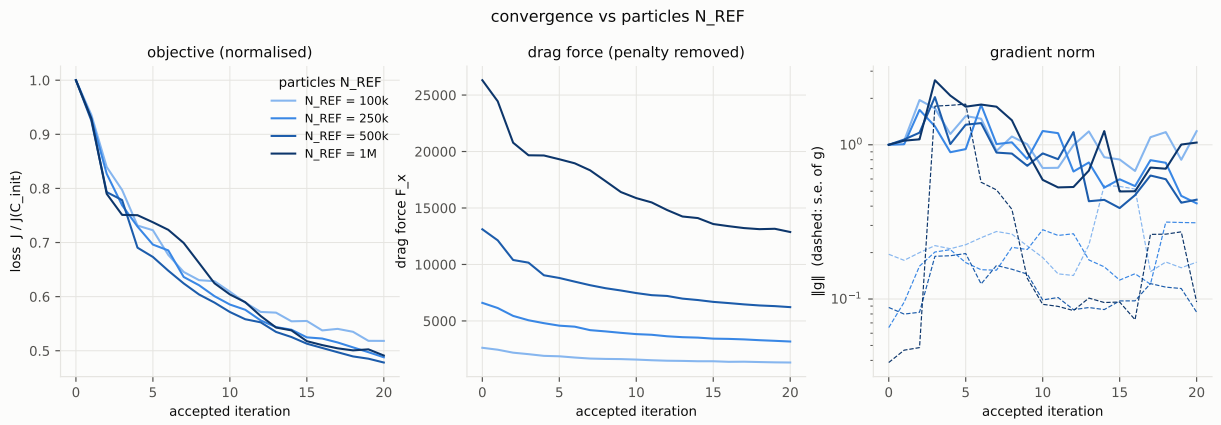


Figure 4: This figure shows the objective and the gradient norm against iteration. The runs differ only in N_REF. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

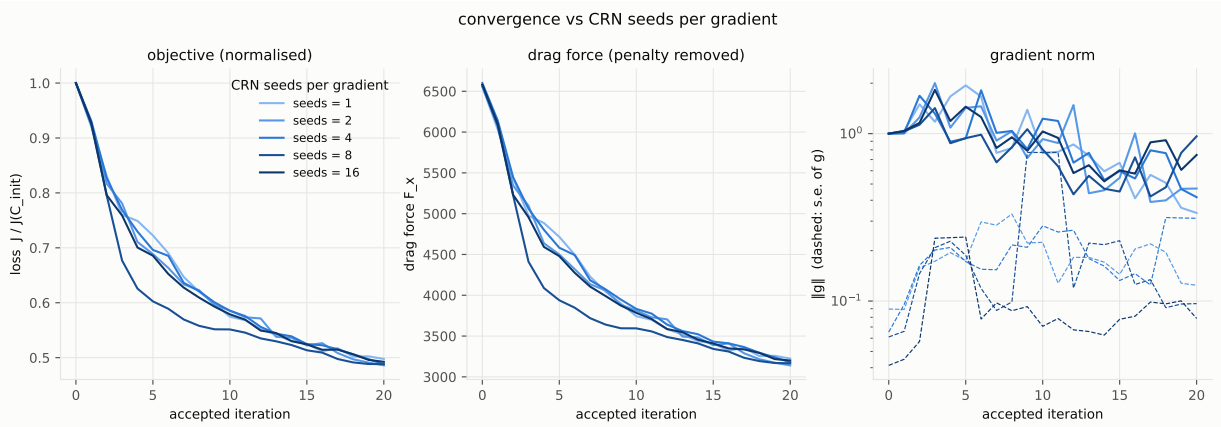


Figure 5: This figure shows the objective and the gradient norm against iteration. The runs differ only in N_AVG. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

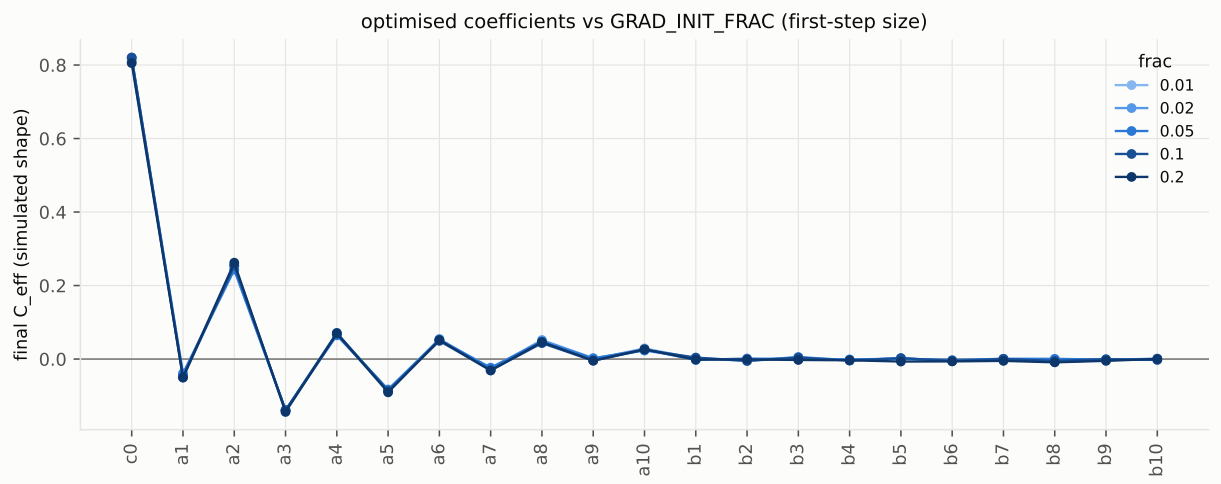


Figure 6: This figure shows the final shape coefficients against GRAD_INIT_FRAC. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

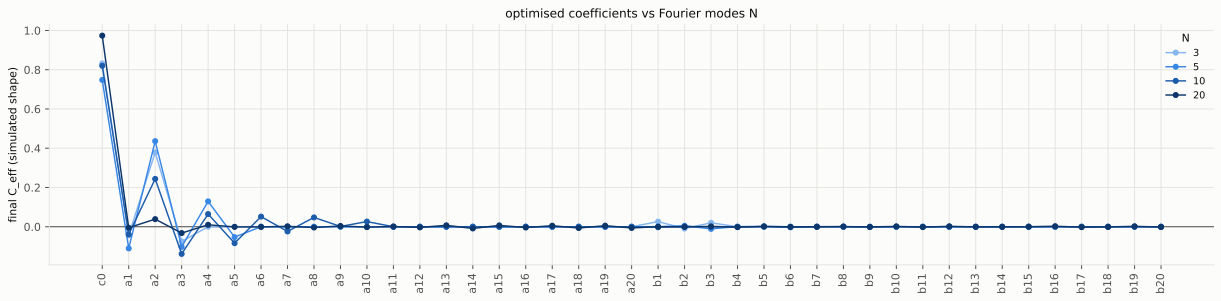


Figure 7: This figure shows the final shape coefficients against N_FOURIER. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

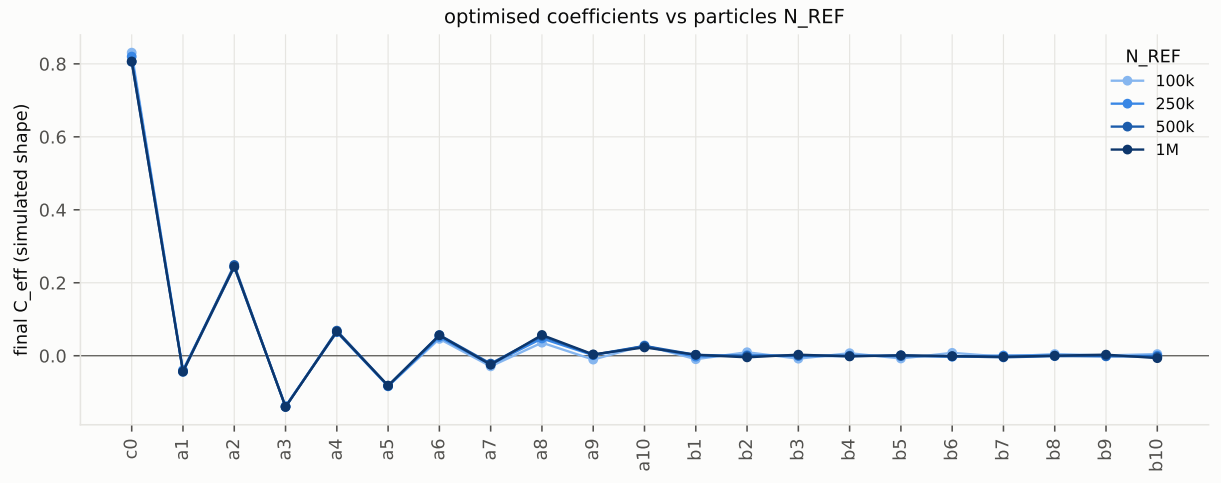


Figure 8: This figure shows the final shape coefficients against N_REF. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

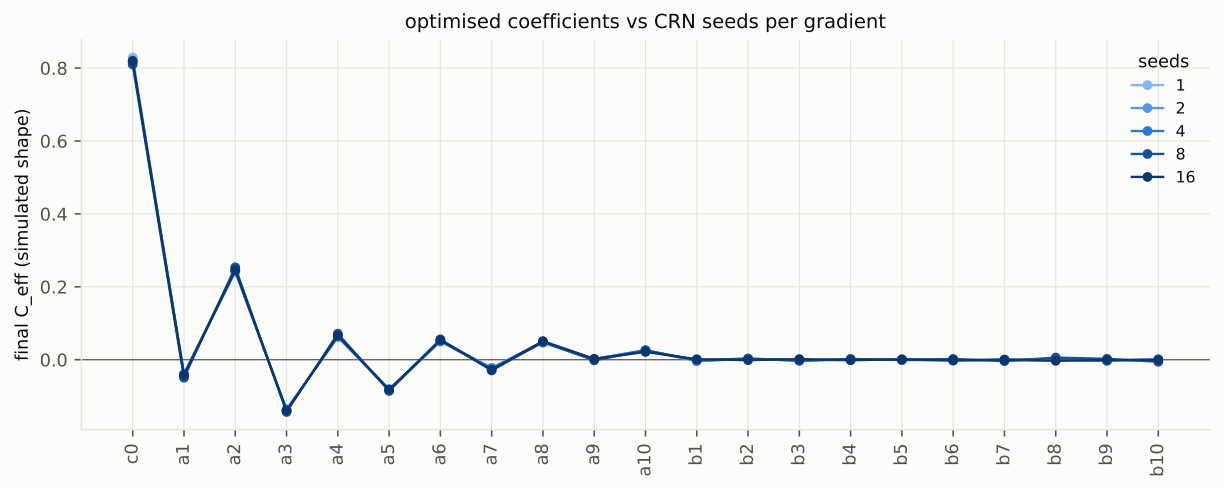


Figure 9: This figure shows the final shape coefficients against N_AVG. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

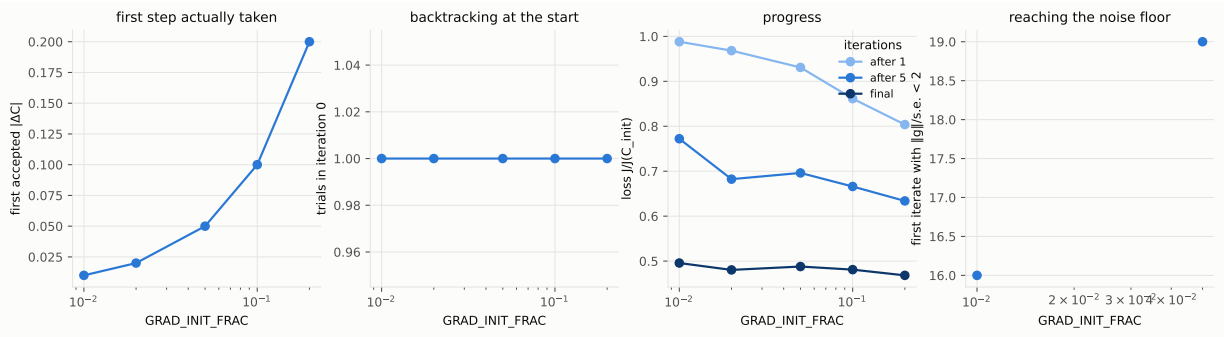


Figure 10: This figure shows how the first-step size affects the first accepted step, the number of backtracks at iteration 0, and the objective after 1, 5 and 20 iterations. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

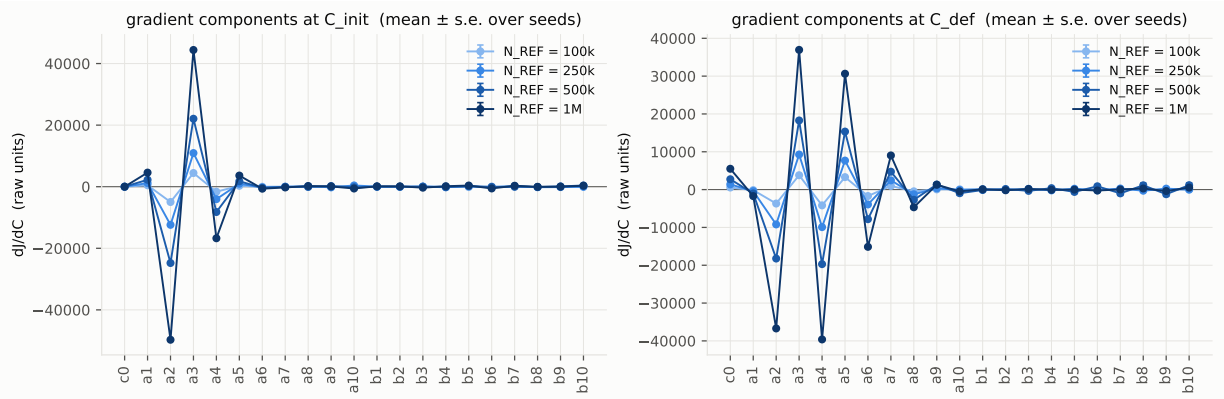


Figure 11: This figure shows the mean gradient of each mode, with its standard error over 16 seeds, against the particle count at the circle and at the stretched shape ($a_2 = 0.15$). Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

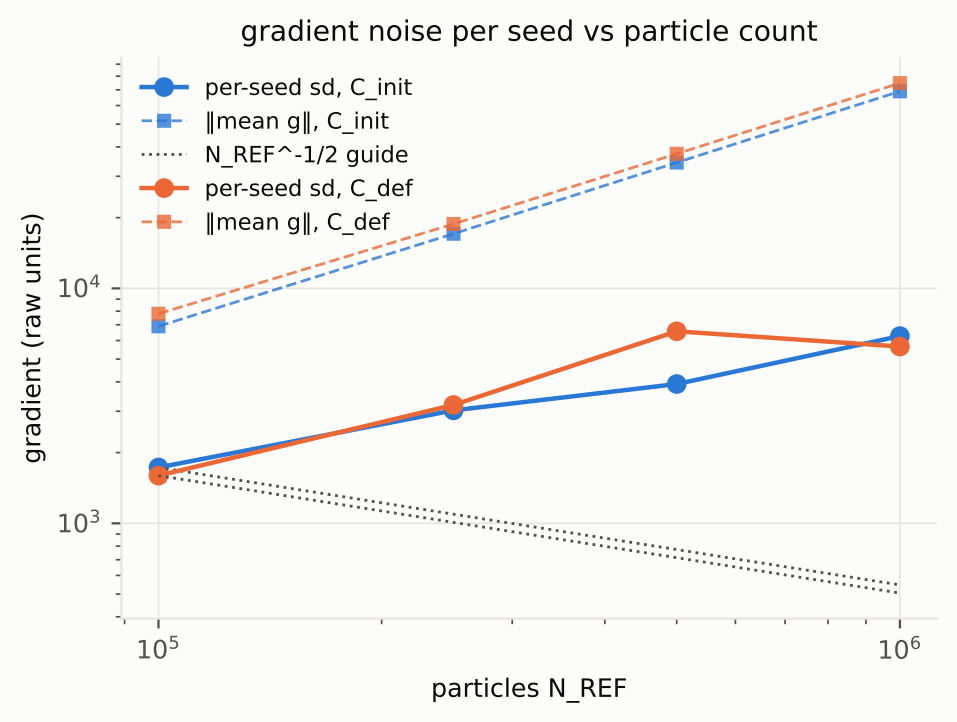


Figure 12: This figure shows the scatter of one seed's gradient against the particle count on logarithmic axes, with an $N^{-1/2}$ guide. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

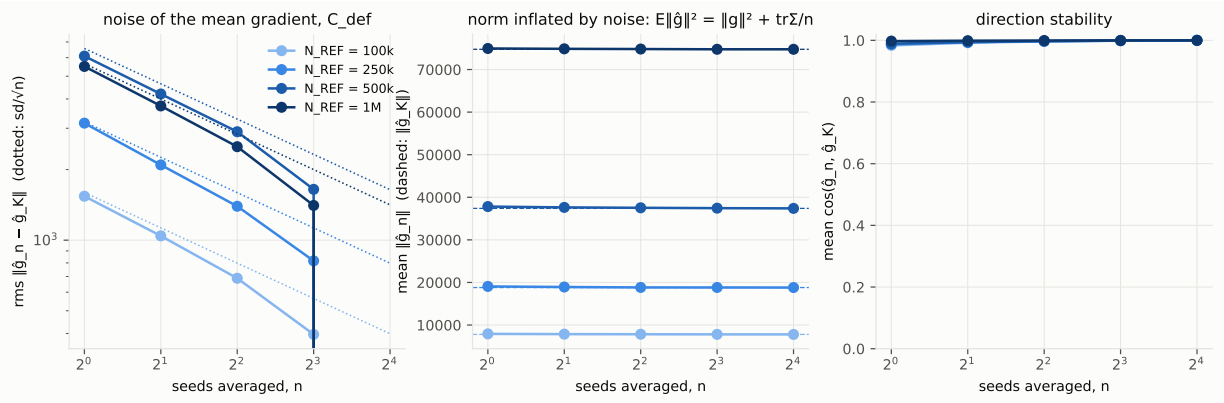


Figure 13: This figure shows the error of an n-seed average against n at the stretched shape ($a_2 = 0.15$), together with the $1/\sqrt{n}$ prediction. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

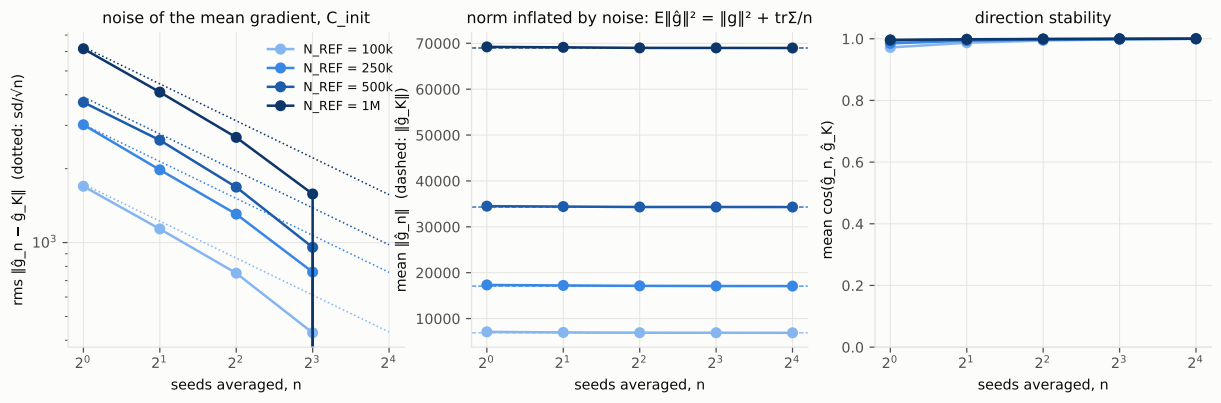


Figure 14: This figure shows the error of an n-seed average against n at the circle, together with the $1/\sqrt{n}$ prediction. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

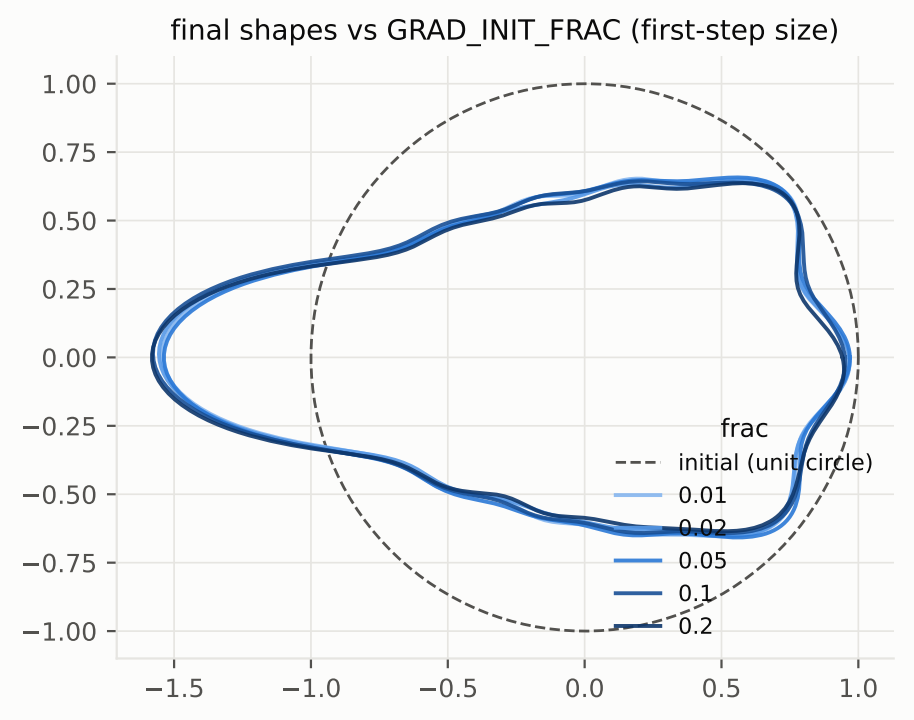


Figure 15: This figure shows the final shapes for each value of GRAD_INIT_FRAC. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

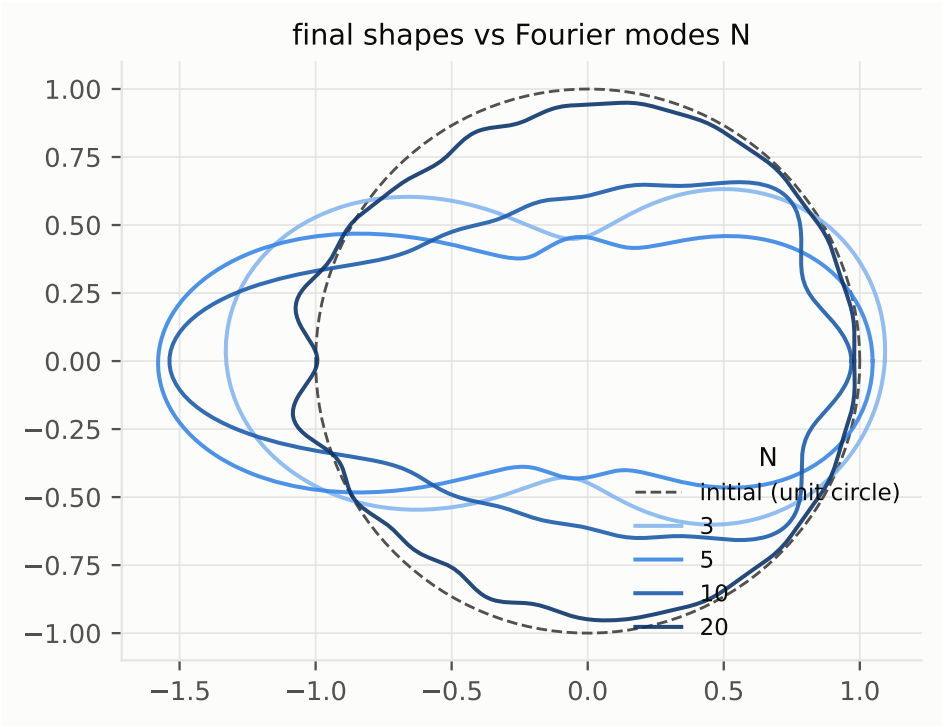


Figure 16: This figure shows the final shapes for each value of N_FOURIER. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

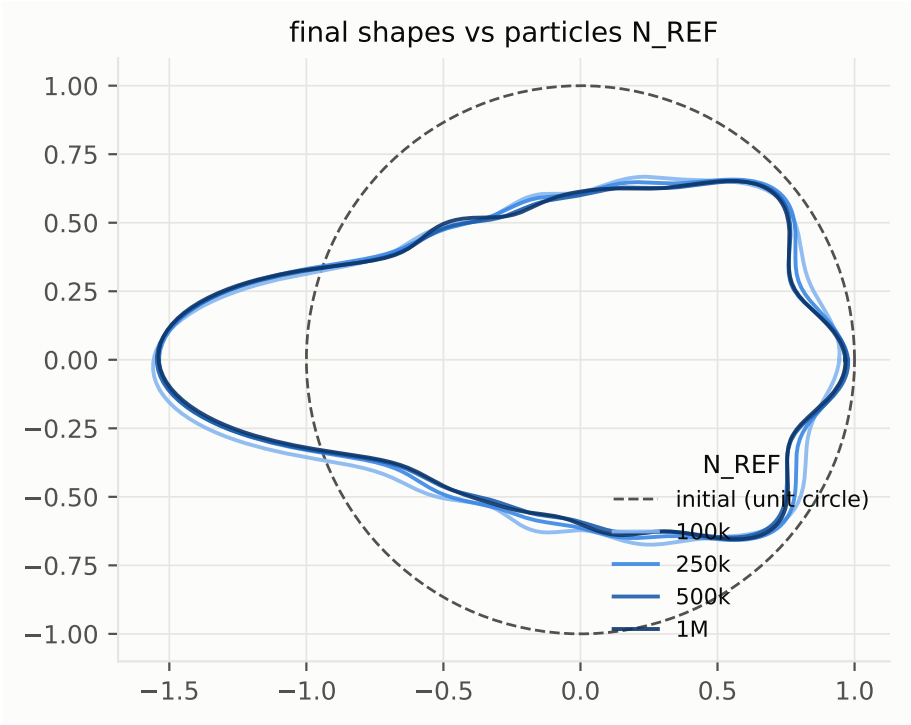


Figure 17: This figure shows the final shapes for each value of N_REF. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

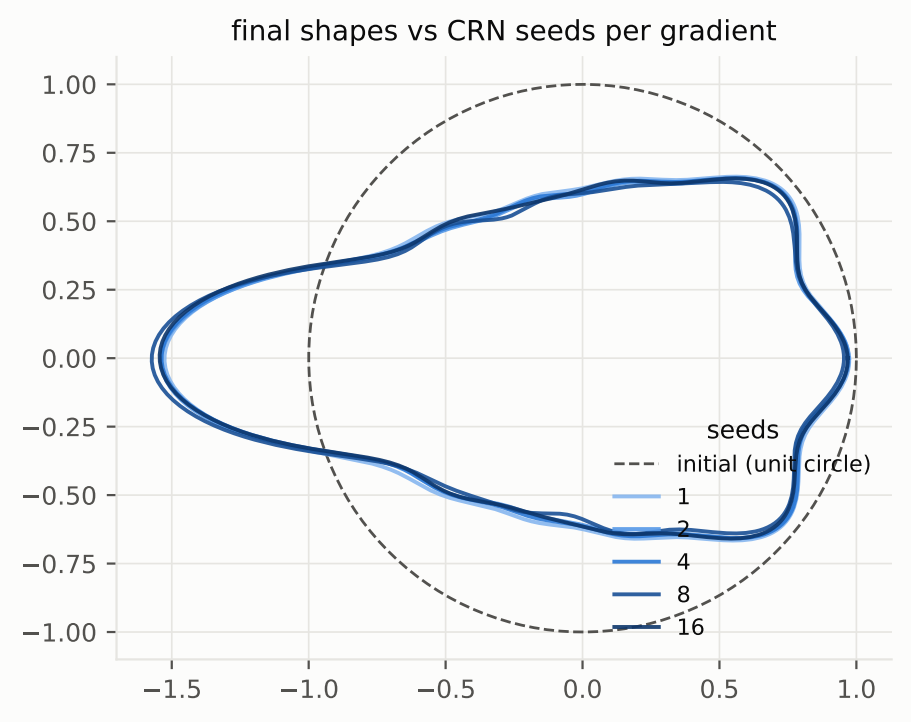


Figure 18: This figure shows the final shapes for each value of N_AVG. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

success classification per run

run	verdict	decayed	plateau	at_noise_floor
baseline/B0_seed0	decaying_noisy	yes	yes	no
sweep_frac/frac0.01	decaying_noisy	yes	no	no
sweep_frac/frac0.02	decaying_noisy	yes	yes	no
sweep_frac/frac0.1	decaying_noisy	yes	no	no
sweep_frac/frac0.2	decaying_noisy	yes	yes	no
sweep_nfourier/nf20	decaying_noisy	yes	no	yes
sweep_nfourier/nf3	decaying_noisy	yes	no	yes
sweep_nfourier/nf5	decaying_noisy	yes	yes	no
sweep_nref/nref100k	decaying_noisy	yes	yes	no
sweep_nref/nref1M	decaying_noisy	yes	yes	no
sweep_nref/nref500k	decaying_noisy	yes	yes	no
sweep_seeds/nseeds1	decaying_noisy	yes	no	n/a
sweep_seeds/nseeds16	decaying_noisy	yes	yes	no
sweep_seeds/nseeds2	decaying_noisy	yes	no	no
sweep_seeds/nseeds8	decaying_noisy	yes	no	no

Figure 19: This figure shows the automatic classifier's verdict for each run. For 1-seed runs a success verdict is an artefact, because a single seed gives no noise estimate. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

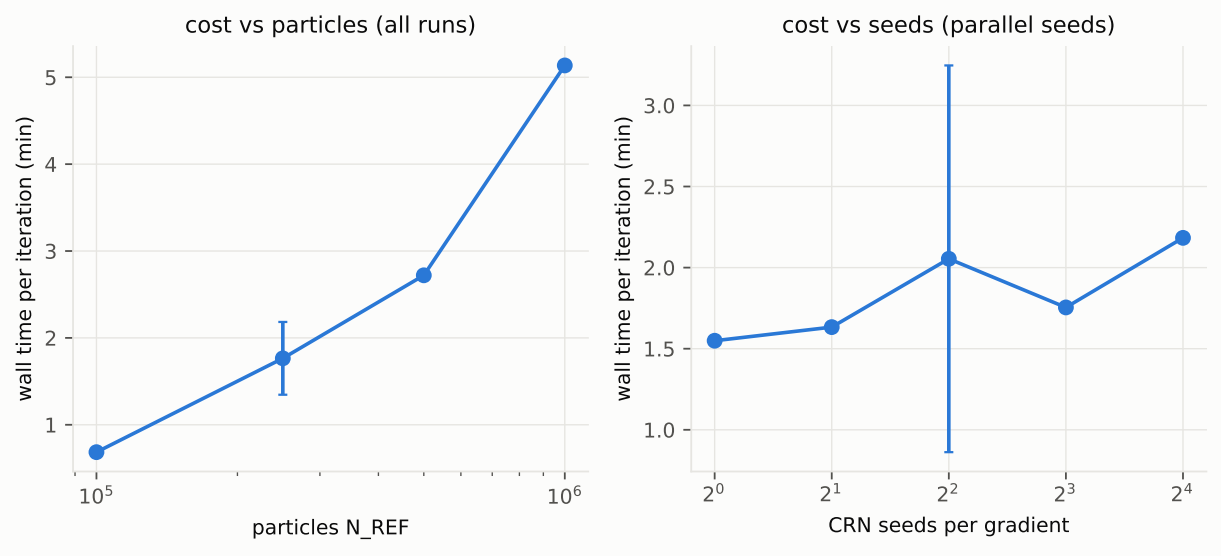


Figure 20: This figure shows the wall-clock time per iteration against the particle count and the number of seeds. Pass 1 used N_REF = 250k, N_AVG = 4, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 20 and SEED0 = 0, except for the parameter that the figure varies.

2.2 Pass 2

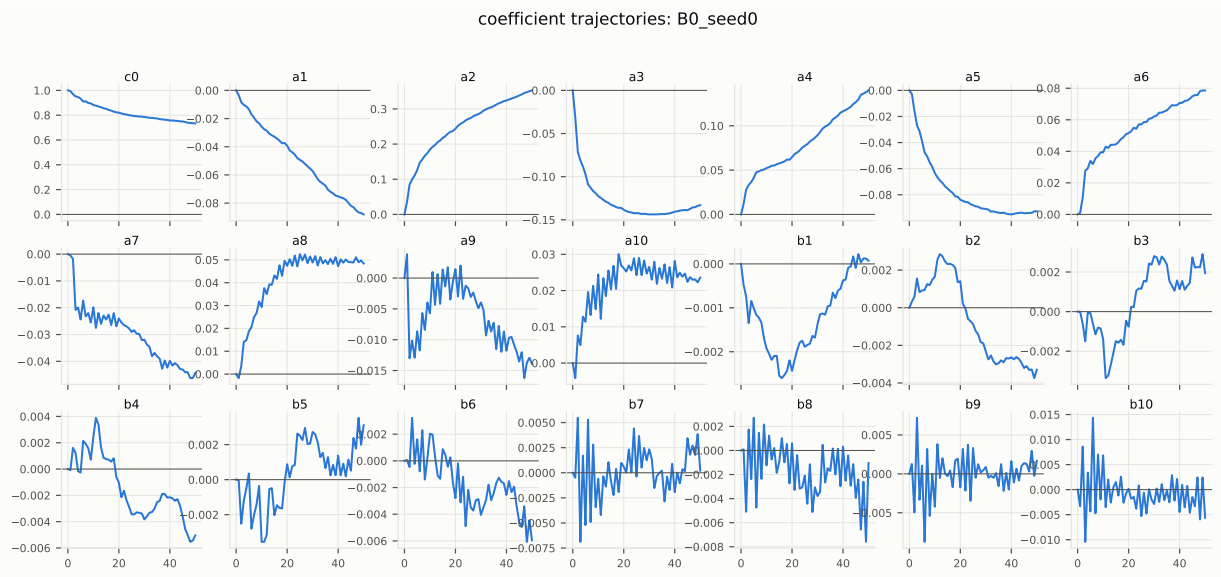


Figure 21: This figure shows each Fourier coefficient of the baseline run with SEED0 = 0 against iteration. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

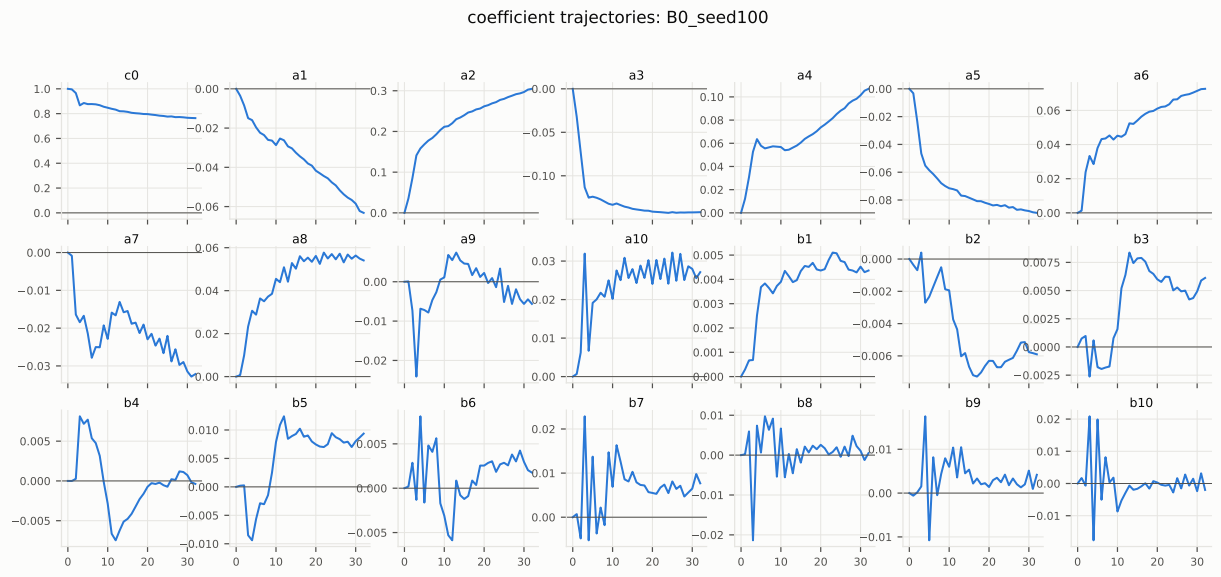


Figure 22: This figure shows each Fourier coefficient of the baseline run with SEED0 = 100 against iteration. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

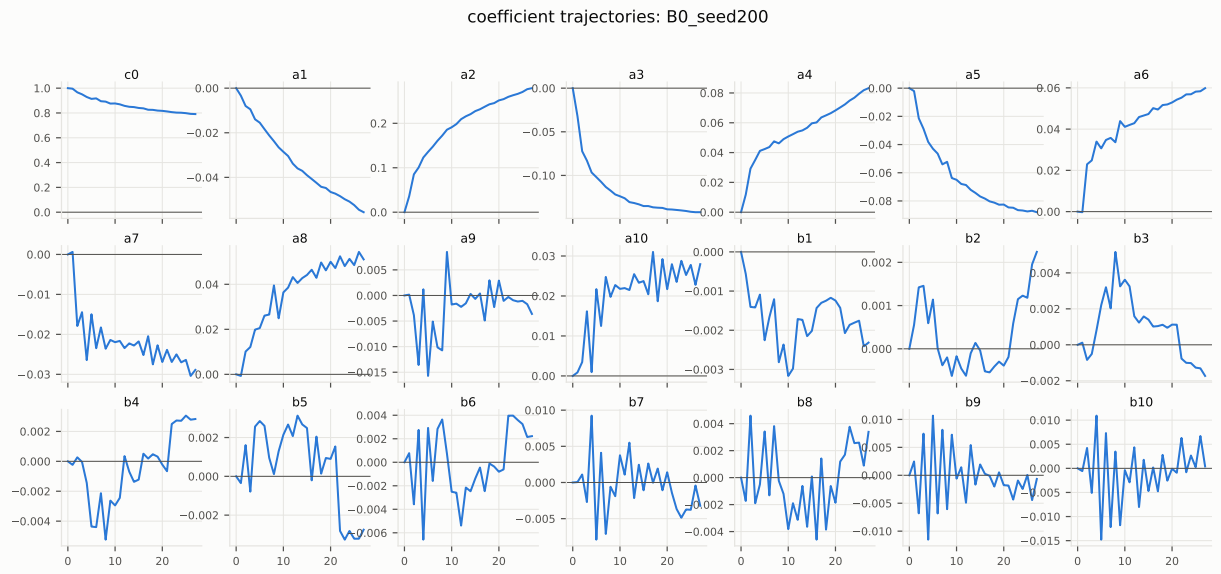


Figure 23: This figure shows each Fourier coefficient of the baseline run with SEED0 = 200 against iteration. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

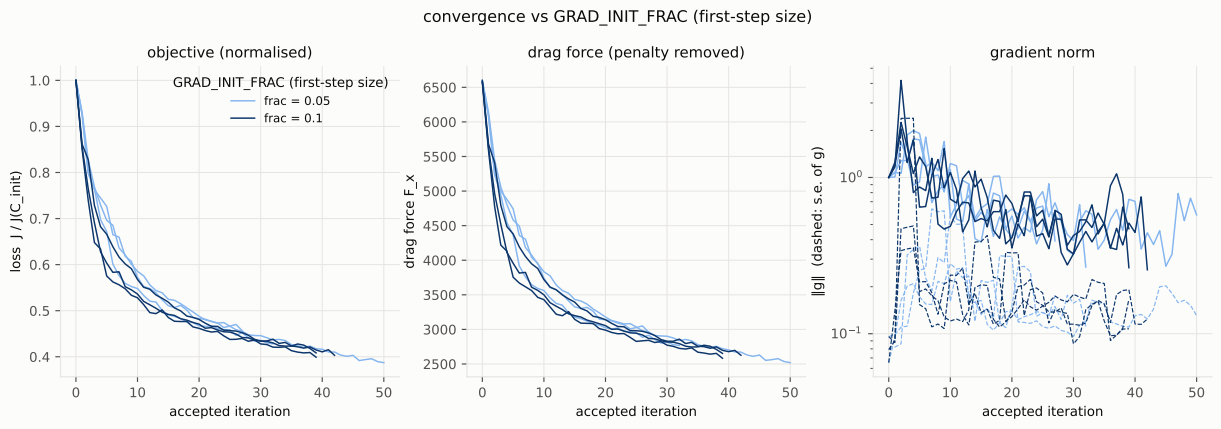


Figure 24: This figure shows the objective and the gradient norm against iteration. The runs differ only in GRAD_INIT_FRAC. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

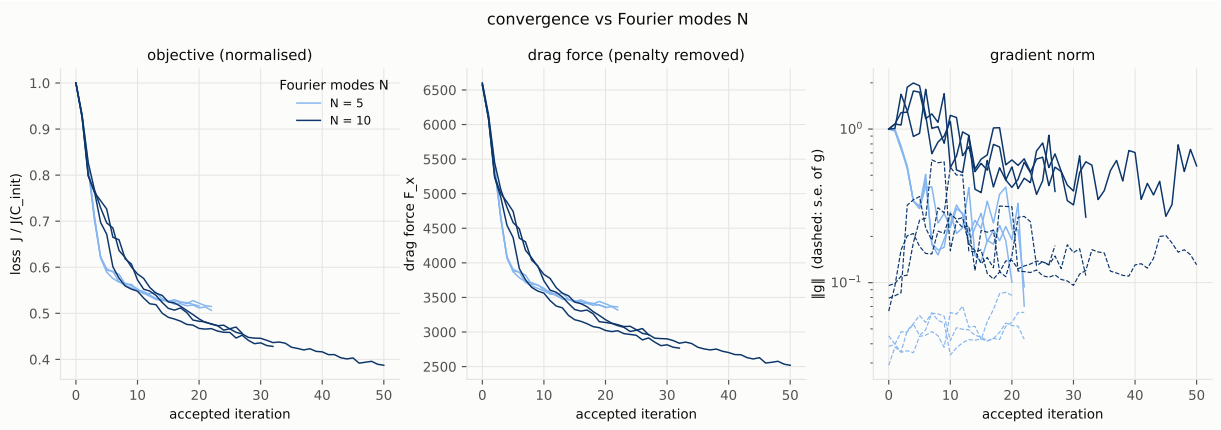


Figure 25: This figure shows the objective and the gradient norm against iteration. The runs differ only in N_FOURIER. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

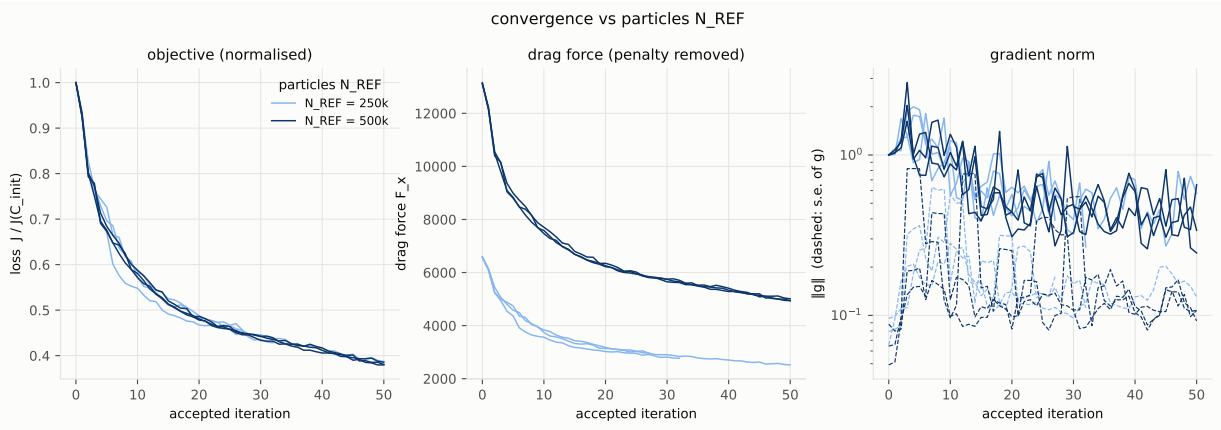


Figure 26: This figure shows the objective and the gradient norm against iteration. The runs differ only in N_REF. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

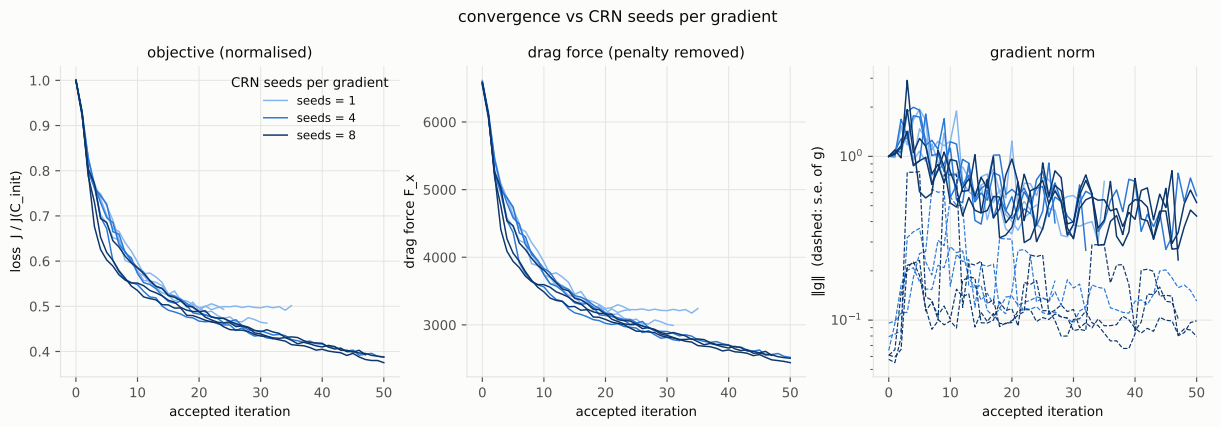


Figure 27: This figure shows the objective and the gradient norm against iteration. The runs differ only in N_AVG. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

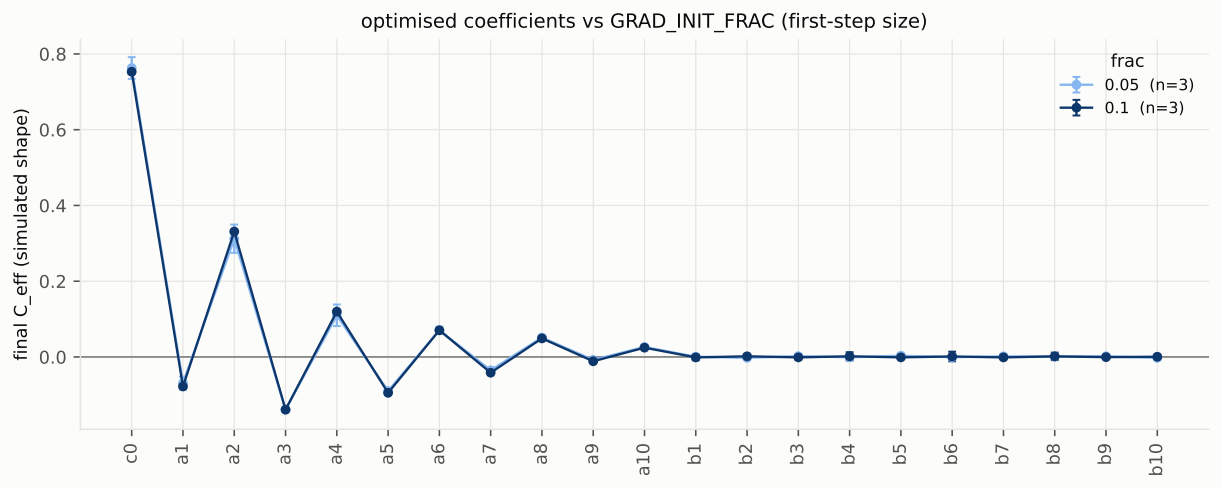


Figure 28: This figure shows the final shape coefficients against GRAD_INIT_FRAC. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

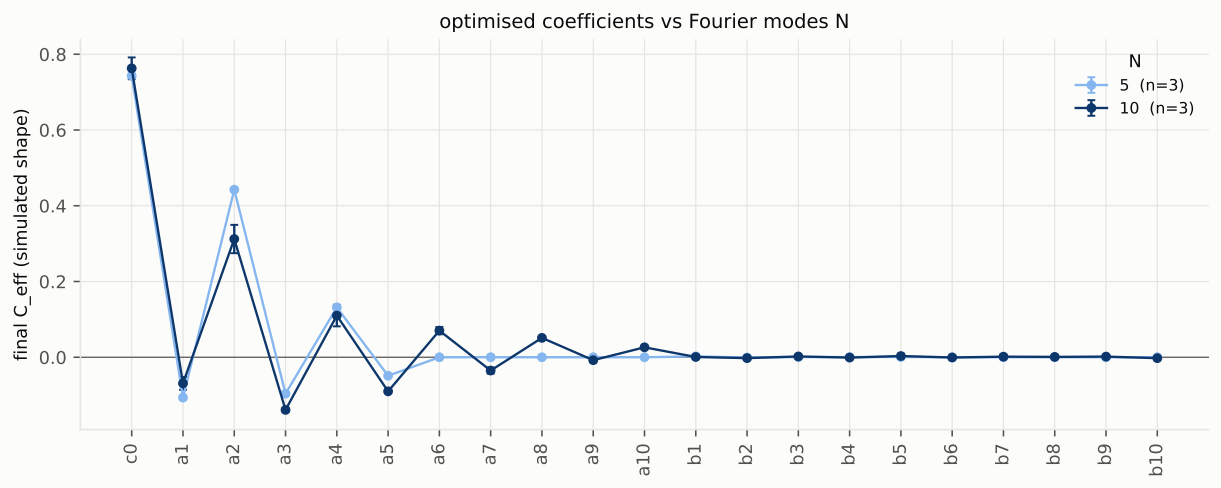


Figure 29: This figure shows the final shape coefficients against N_FOURIER. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

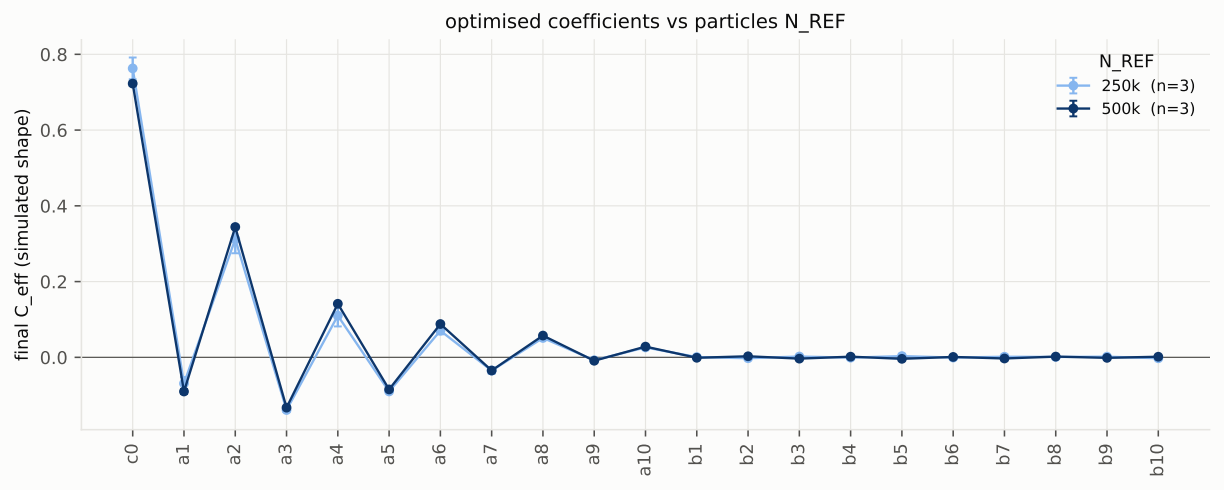


Figure 30: This figure shows the final shape coefficients against N_REF. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

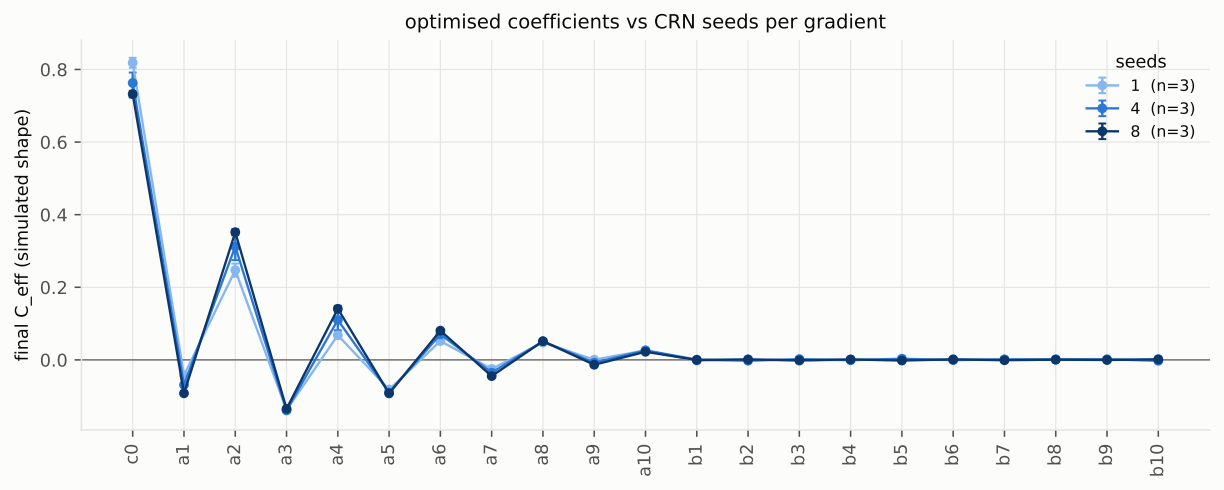


Figure 31: This figure shows the final shape coefficients against N_AVG. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

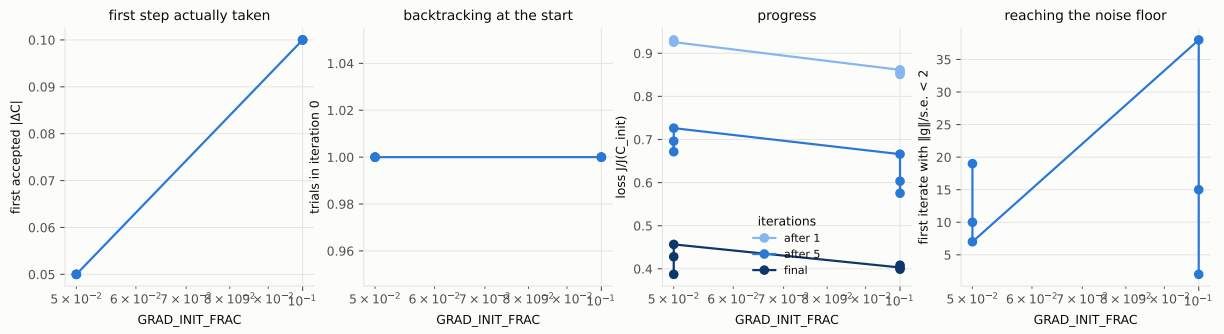


Figure 32: This figure shows how the first-step size affects the first accepted step, the number of backtracks at iteration 0, and the objective after 1, 5 and 20 iterations. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

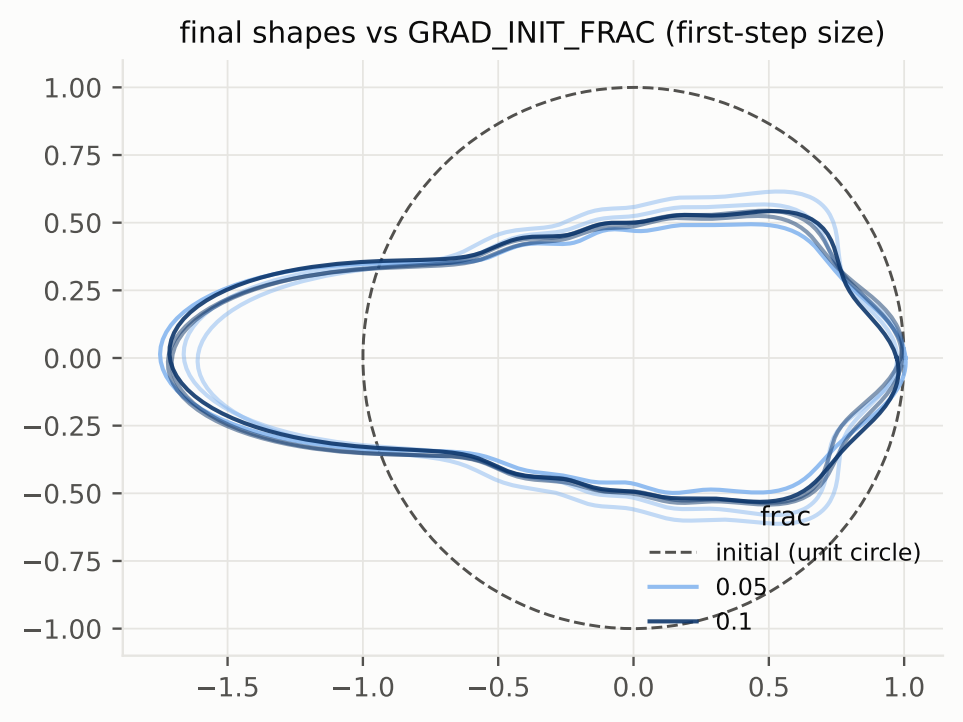


Figure 33: This figure shows the final shapes for each value of GRAD_INIT_FRAC. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

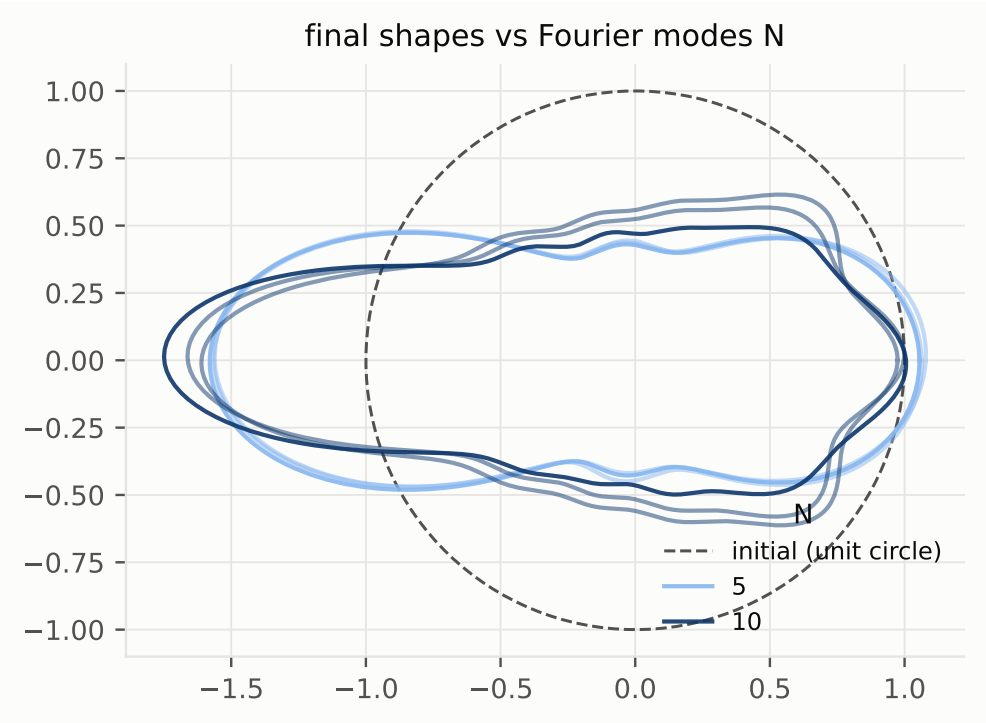


Figure 34: This figure shows the final shapes for each value of N_FOURIER. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

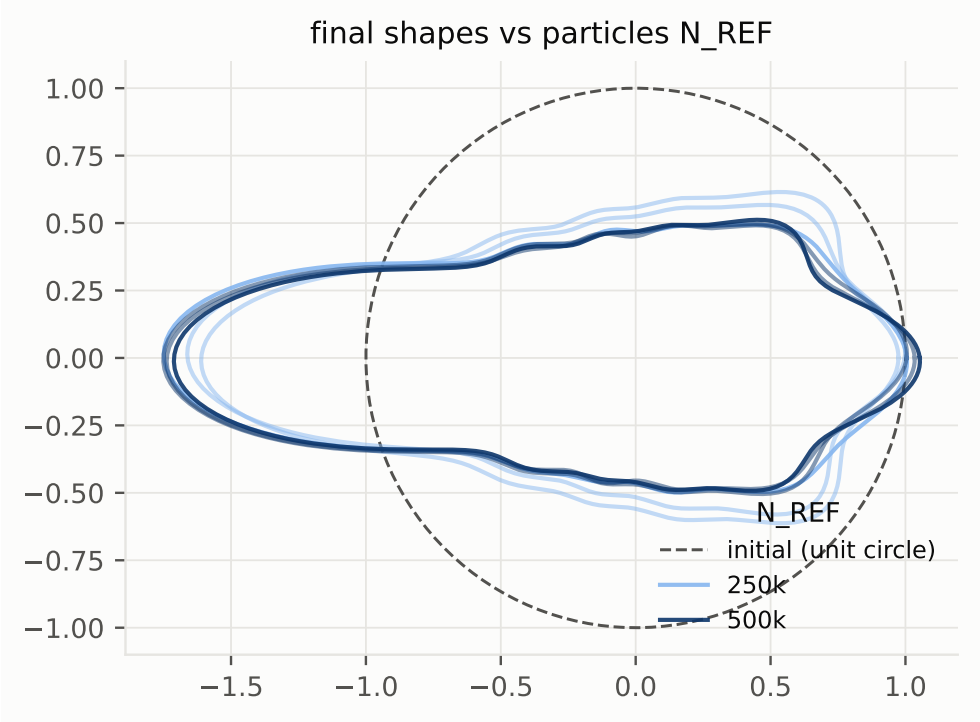


Figure 35: This figure shows the final shapes for each value of N_REF. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

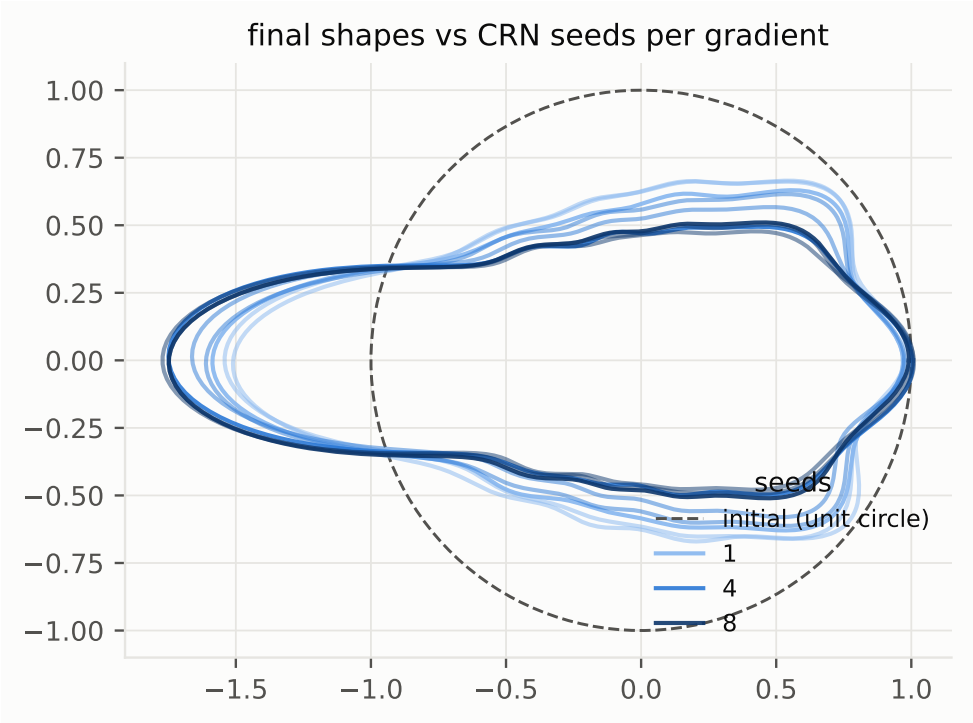


Figure 36: This figure shows the final shapes for each value of N_AVG. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

success classification per run

run	verdict	decayed	plateau	at_noise_floor
baseline/B0_seed0	decaying_noisy	yes	yes	no
baseline/B0_seed100	decaying_noisy	yes	yes	no
baseline/B0_seed200	decaying_noisy	yes	yes	no
sweep_frac/frac0.1_seed0	decaying_noisy	yes	yes	no
sweep_frac/frac0.1_seed100	decaying_noisy	yes	yes	no
sweep_frac/frac0.1_seed200	decaying_noisy	yes	no	no
sweep_nfourier/nf5_seed0	decaying_noisy	yes	no	no
sweep_nfourier/nf5_seed100	decaying_noisy	yes	yes	no
sweep_nfourier/nf5_seed200	decaying_noisy	yes	no	no
sweep_nref/nref500k_seed0	decaying_noisy	yes	yes	no
sweep_nref/nref500k_seed100	decaying_noisy	yes	yes	no
sweep_nref/nref500k_seed200	decaying_noisy	yes	no	no
sweep_seeds/nseeds1_seed0	success	yes	yes	n/a
sweep_seeds/nseeds1_seed100	success	yes	yes	n/a
sweep_seeds/nseeds1_seed200	decaying_noisy	yes	no	n/a
sweep_seeds/nseeds8_seed0	decaying_noisy	yes	yes	no
sweep_seeds/nseeds8_seed100	decaying_noisy	yes	yes	no
sweep_seeds/nseeds8_seed200	decaying_noisy	yes	yes	no

Figure 37: This figure shows the automatic classifier's verdict for each run. For 1-seed runs a success verdict is an artefact, because a single seed gives no noise estimate. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

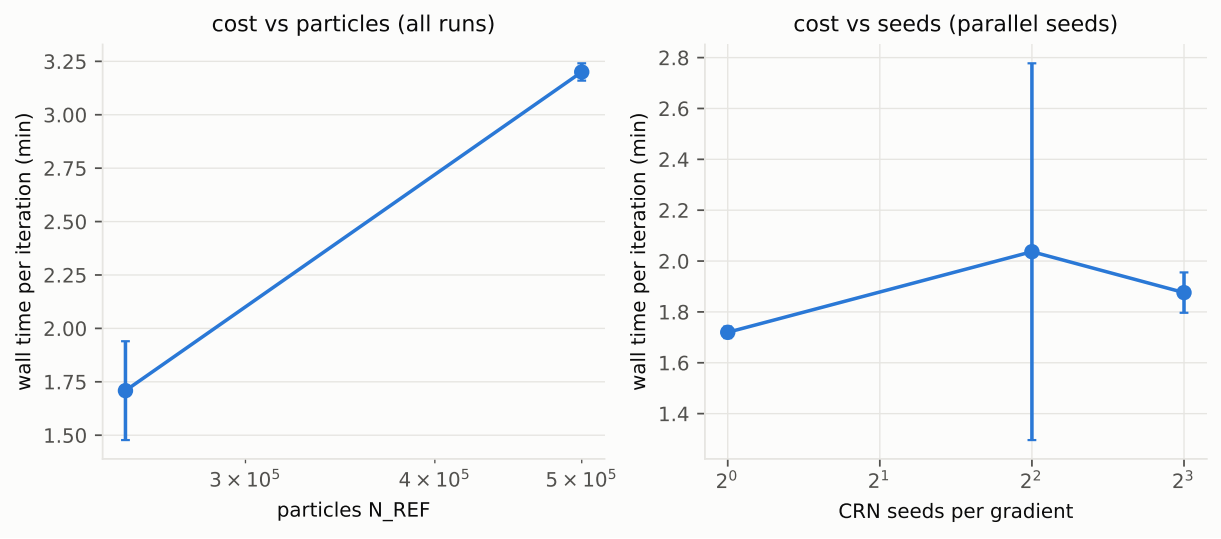


Figure 38: This figure shows the wall-clock time per iteration against the particle count and the number of seeds. Pass 2 used the same settings with N_ITER = 50 and SEED0 = 0, 100 and 200.

2.3 Pass 3

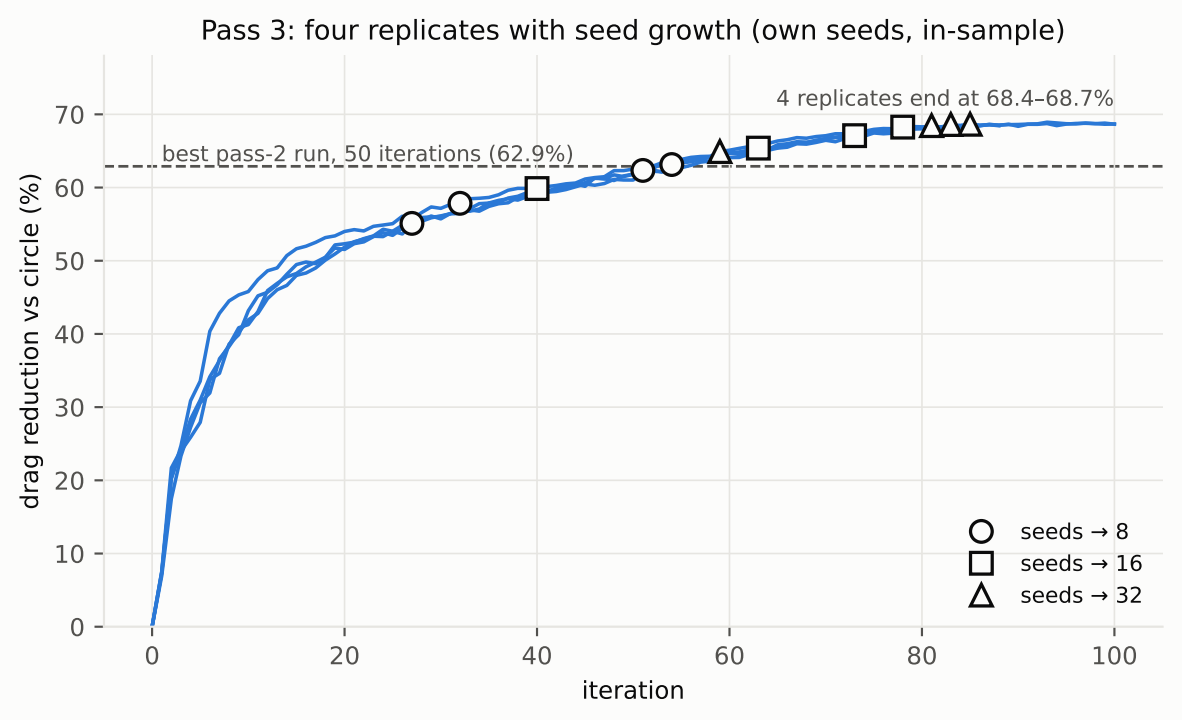


Figure 39: This figure shows the drag reduction against iteration for the four pass-3 replicates on their own seeds. The markers show where each run doubled its seeds instead of stopping. Pass 3 used N_REF = 250k, N_AVG = 4, N_AVG_MAX = 32, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 100 and SEED0 = 0, 100, 200 and 300.

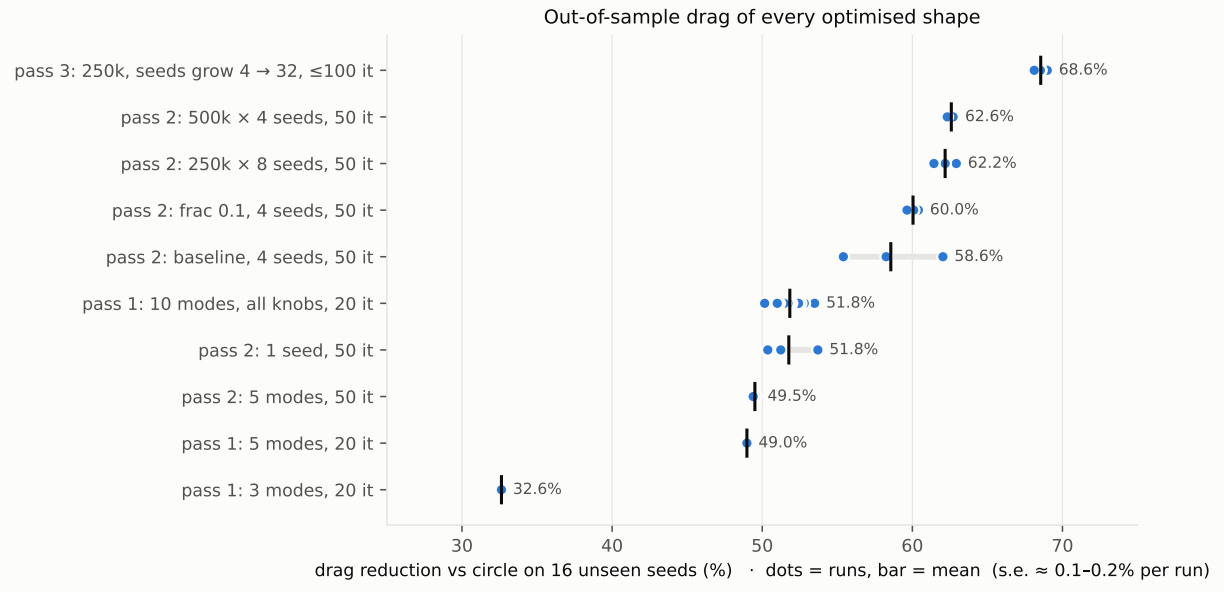


Figure 40: This figure shows the drag reduction of every optimised shape on 16 unseen seeds, grouped by setting. Each dot is one run, and the bar marks the mean. Pass 3 used N_REF = 250k, N_AVG = 4, N_AVG_MAX = 32, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 100 and SEED0 = 0, 100, 200 and 300.

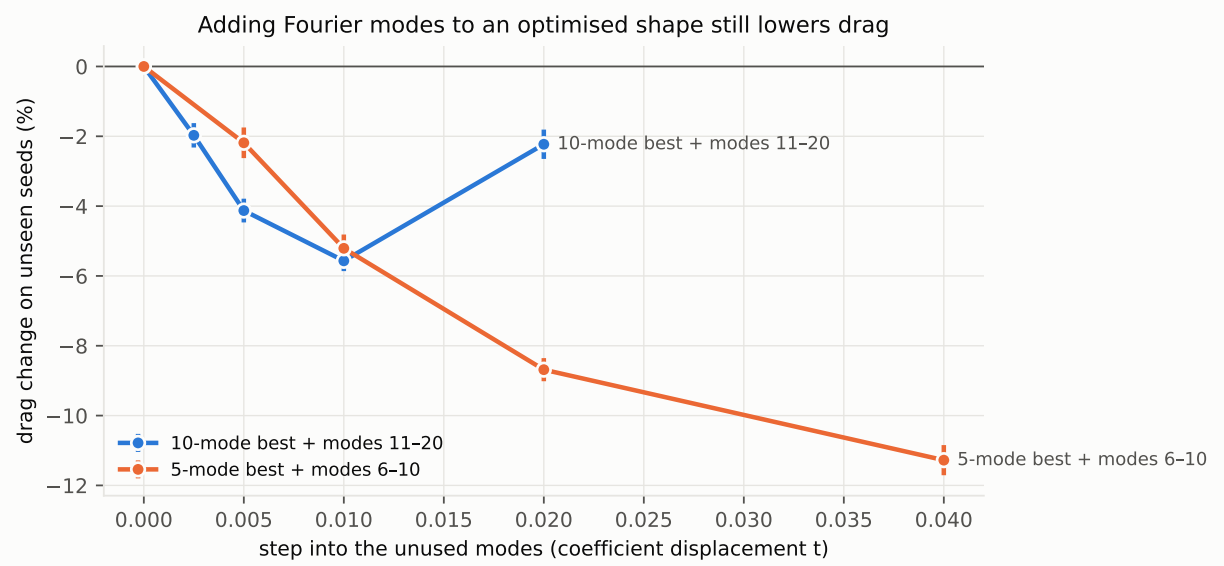


Figure 41: This figure shows how the drag changes when the optimised shapes are stepped into the modes they did not use. The drag was re-measured on unseen seeds, and the error bars are ±1 standard error. Pass 3 used N_REF = 250k, N_AVG = 4, N_AVG_MAX = 32, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 100 and SEED0 = 0, 100, 200 and 300.

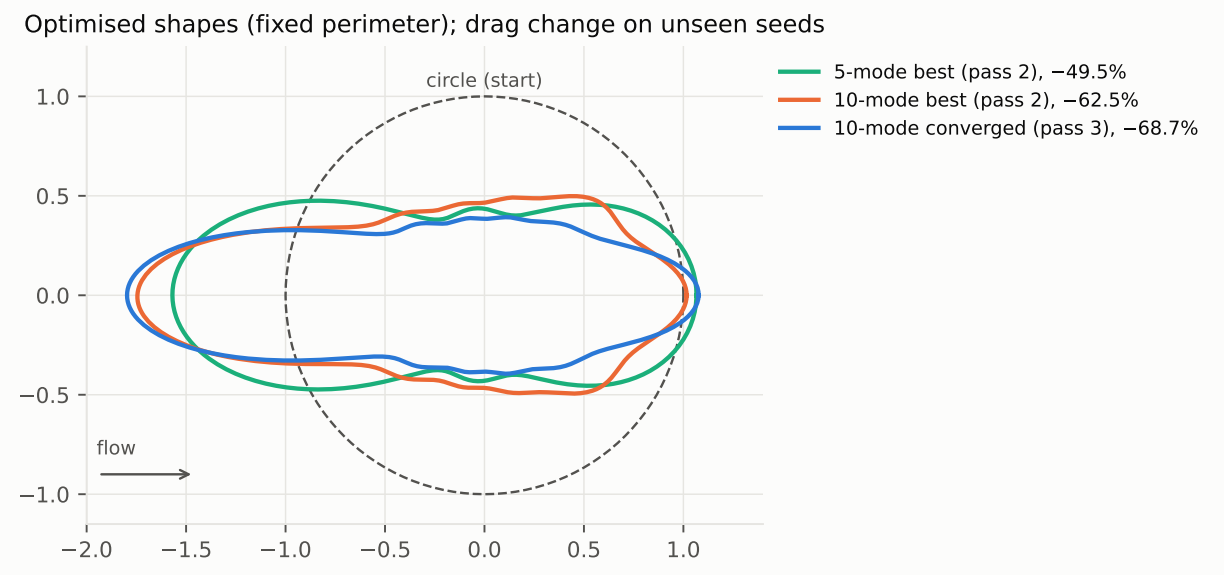


Figure 42: This figure shows the best 5-mode and 10-mode shapes from pass 2 and the final 10-mode shape from pass 3, with their drag reductions on unseen seeds. All shapes have the same perimeter, and the gas flows from left to right. Pass 3 used N_REF = 250k, N_AVG = 4, N_AVG_MAX = 32, N_FOURIER = 10, GRAD_INIT_FRAC = 0.05, N_ITER = 100 and SEED0 = 0, 100, 200 and 300.

3 Every run

This section shows each of the 38 optimisation runs. Each figure shows the drag reduction against iteration, with markers where the seed count doubled, the gradient norm with its standard error, and the final shape. The parameter values of each run are printed above its figure, and every other setting was the same in every run.

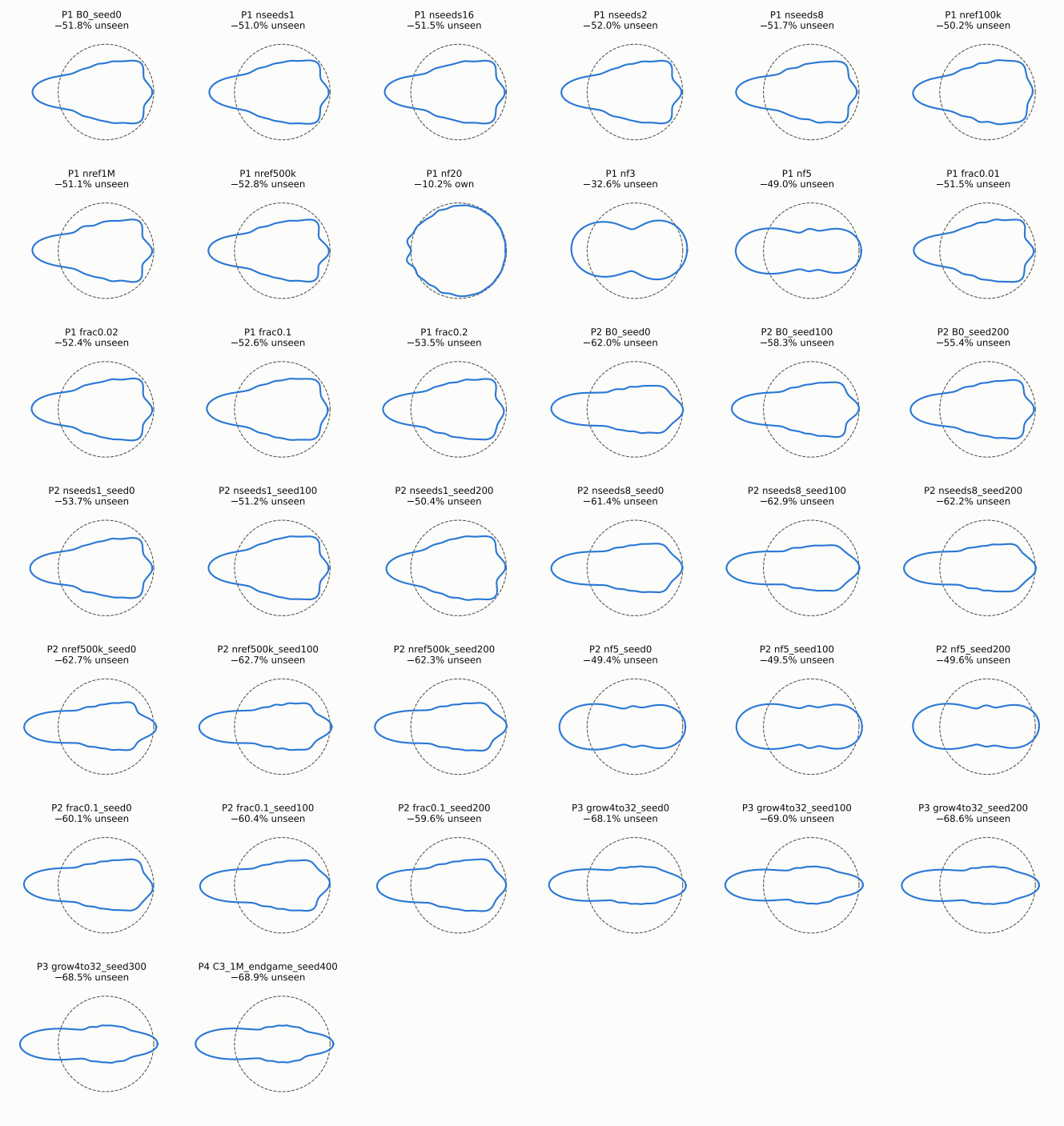


Figure 43: This figure shows every final shape, with the circle dashed. Each label gives the pass, the run and the drag reduction on unseen seeds, or on its own seeds where the shape was not re-measured.

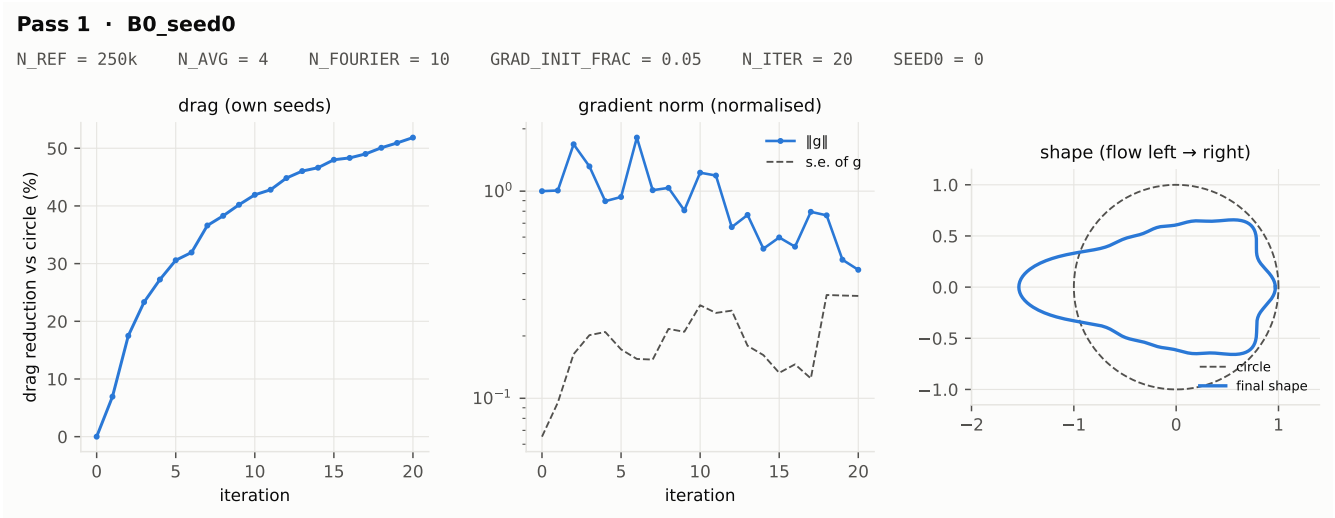


Figure 44: Pass 1, `B0_seed0`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 51.9% relative to the circle on its own seeds and 51.8% relative to the circle on unseen seeds. It took 0.6 hours.

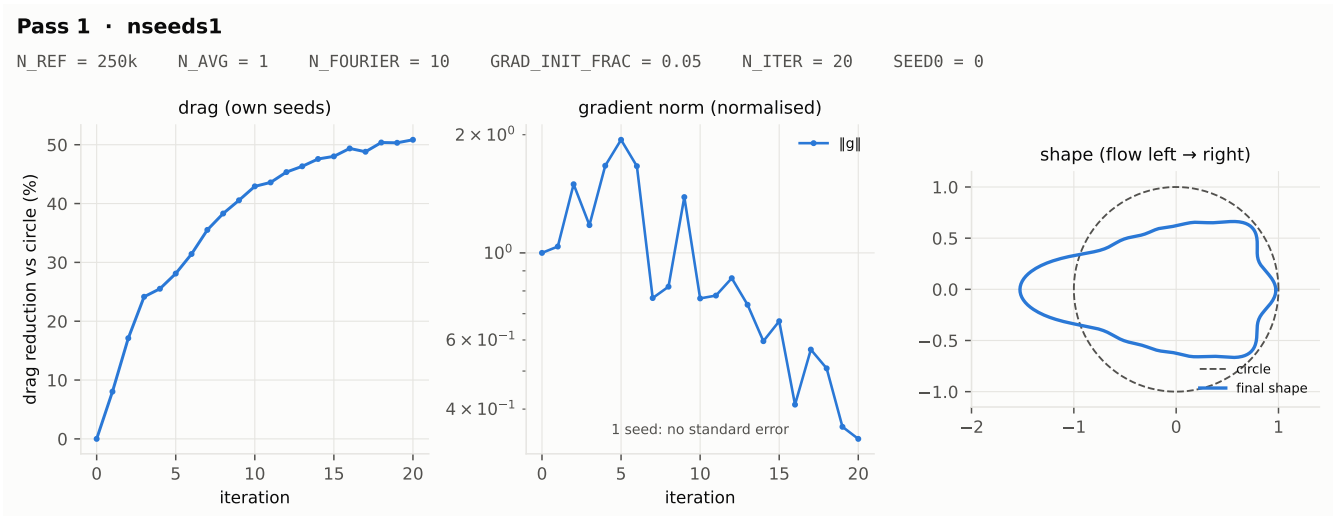


Figure 45: Pass 1, `nseeds1`. The run stopped at the iteration limit after 20 iterations with 1 seeds. Its drag reduction was 50.8% relative to the circle on its own seeds and 51.0% relative to the circle on unseen seeds. It took 0.5 hours.

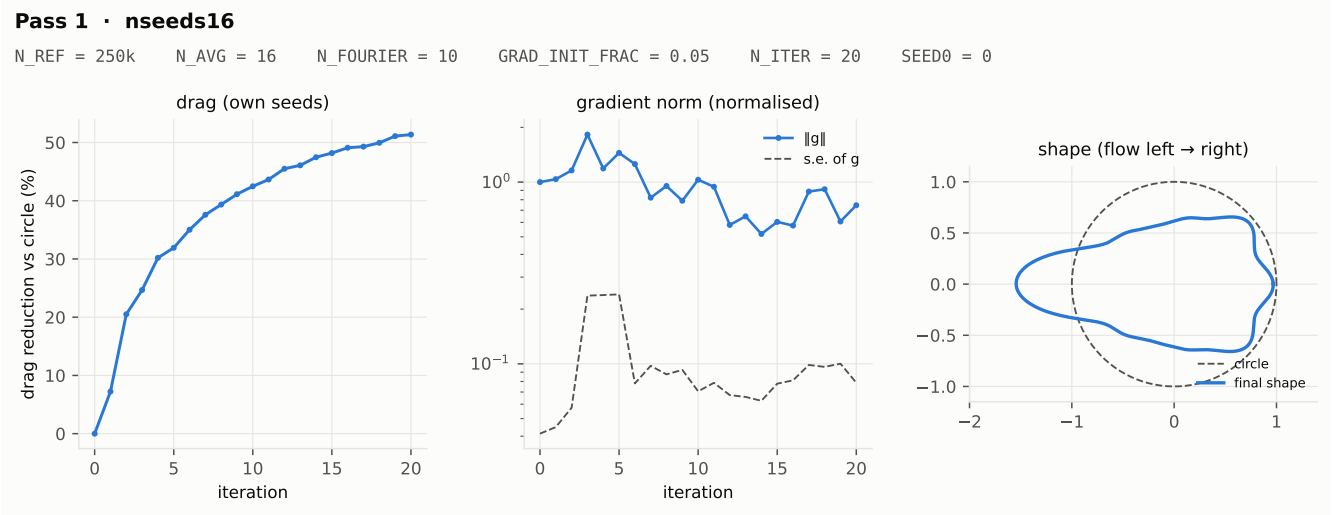


Figure 46: Pass 1, `nseeds16`. The run stopped at the iteration limit after 20 iterations with 16 seeds. Its drag reduction was 51.4% relative to the circle on its own seeds and 51.5% relative to the circle on unseen seeds. It took 0.8 hours.

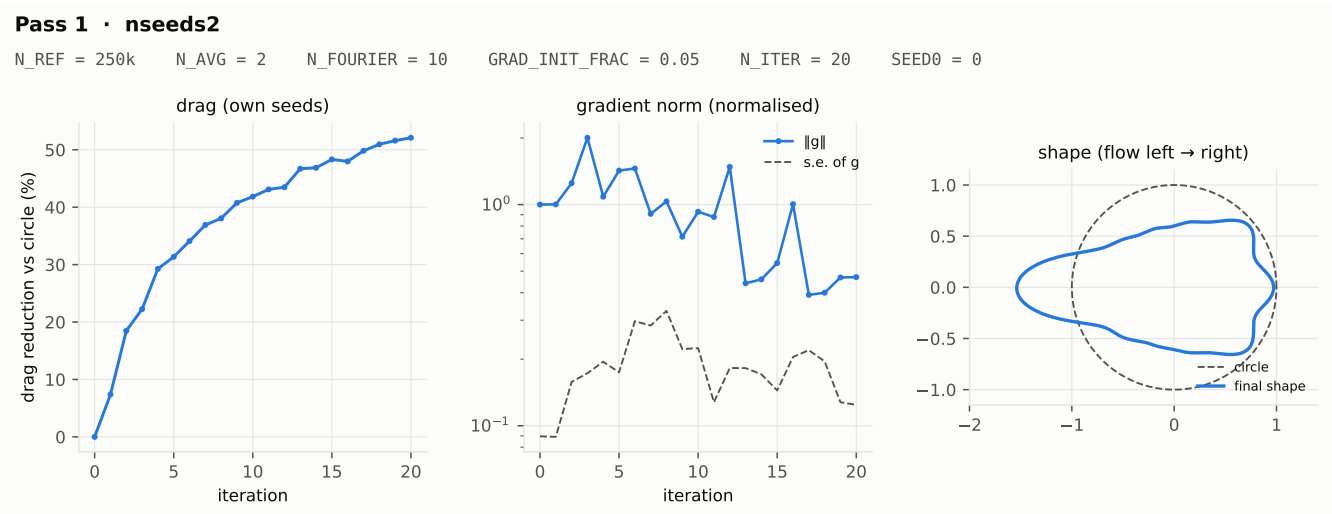


Figure 47: Pass 1, `nseeds2`. The run stopped at the iteration limit after 20 iterations with 2 seeds. Its drag reduction was 52.1% relative to the circle on its own seeds and 52.0% relative to the circle on unseen seeds. It took 0.6 hours.

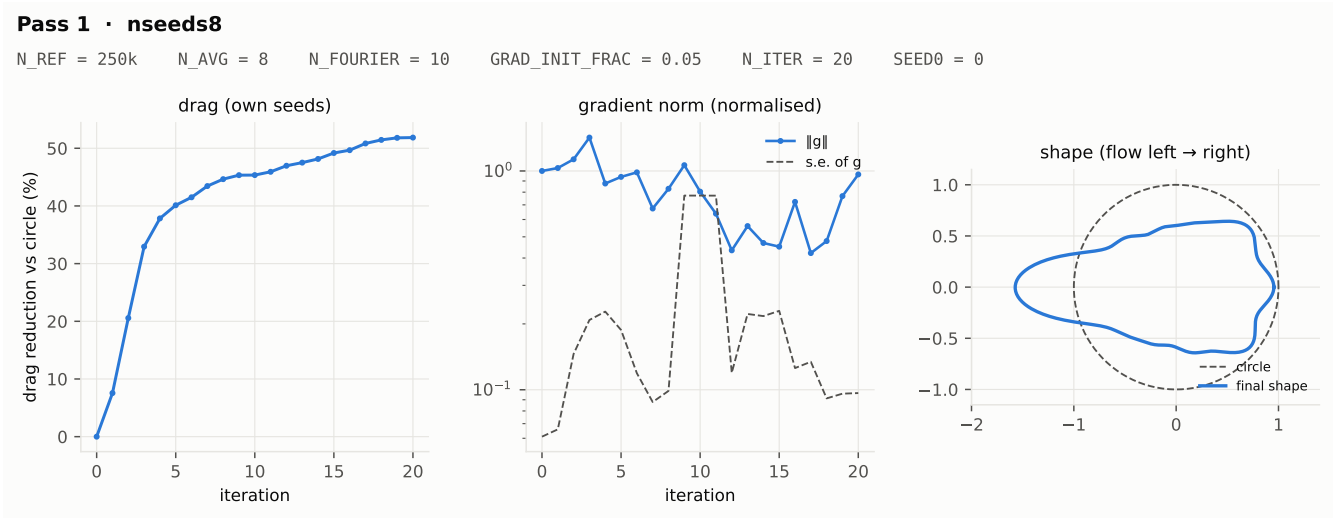


Figure 48: Pass 1, `nseeds8`. The run stopped at the iteration limit after 20 iterations with 8 seeds. Its drag reduction was 51.9% relative to the circle on its own seeds and 51.7% relative to the circle on unseen seeds. It took 0.6 hours.

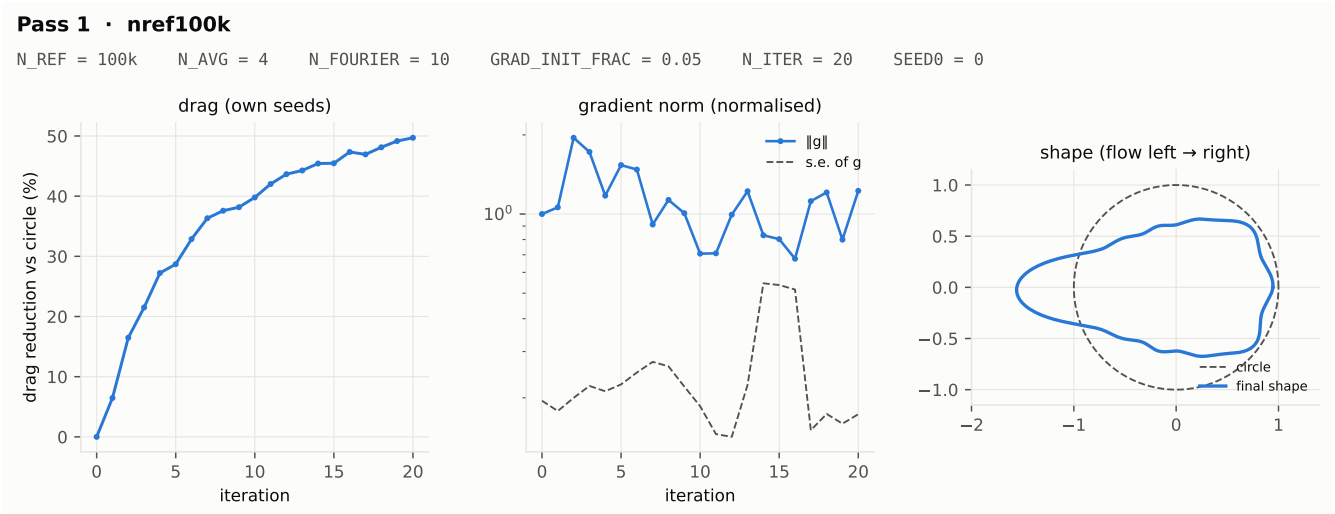


Figure 49: Pass 1, `nref100k`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 49.7% relative to the circle on its own seeds and 50.2% relative to the circle on unseen seeds. It took 0.2 hours.

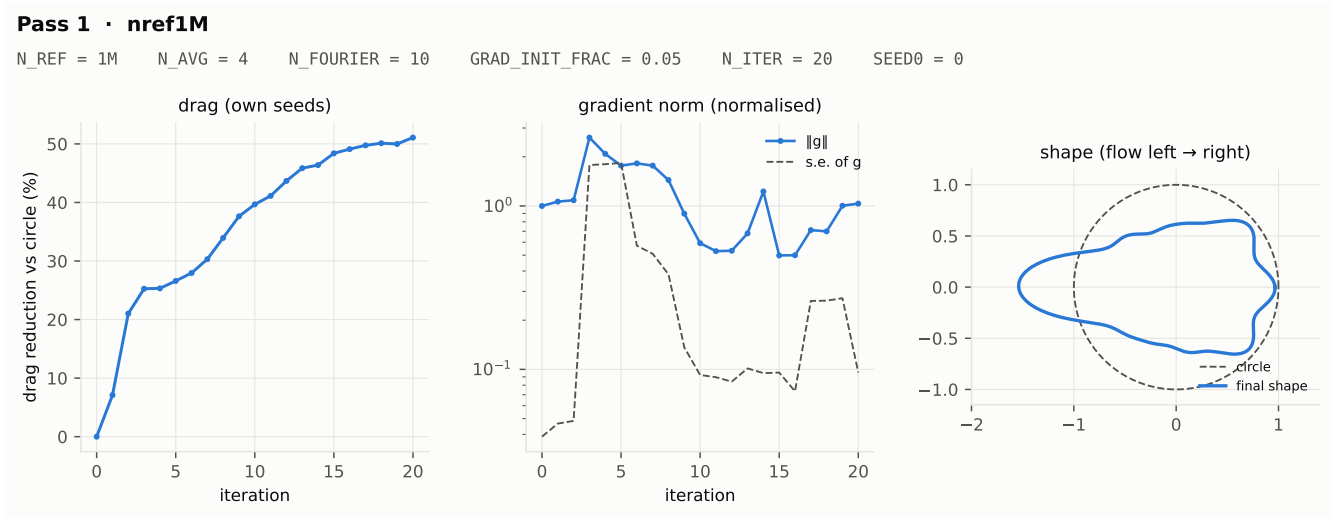


Figure 50: Pass 1, `nref1M`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 51.1% relative to the circle on its own seeds and 51.1% relative to the circle on unseen seeds. It took 1.8 hours.

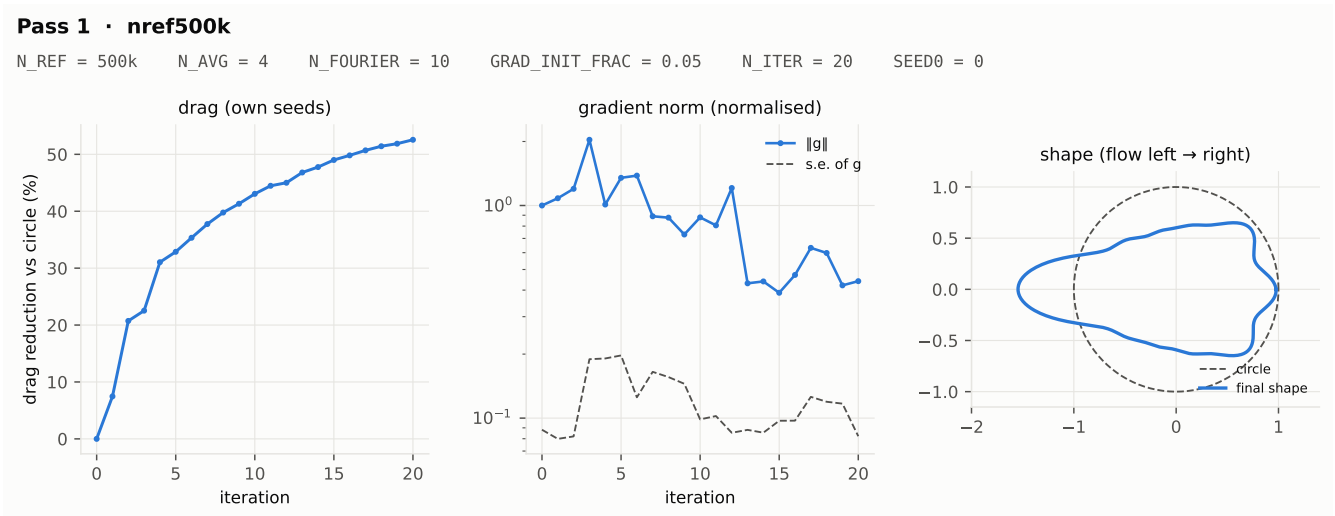


Figure 51: Pass 1, `nref500k`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 52.6% relative to the circle on its own seeds and 52.8% relative to the circle on unseen seeds. It took 0.9 hours.

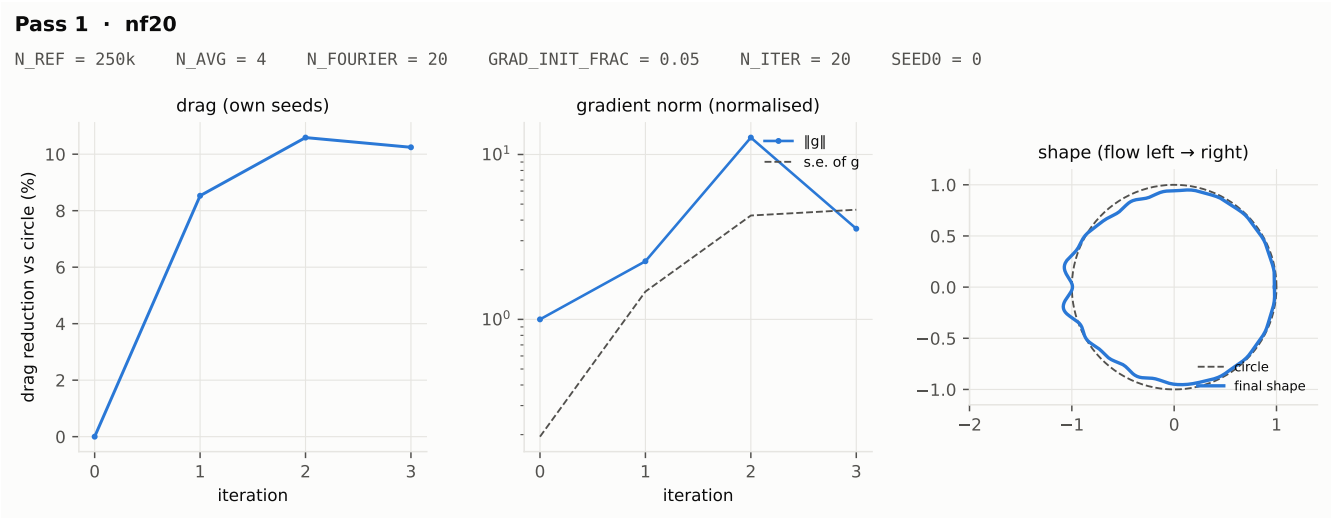


Figure 52: Pass 1, `nf20`. The run stopped at the noise limit after 3 iterations with 4 seeds. Its drag reduction was 10.2% relative to the circle on its own seeds. It took 0.2 hours.

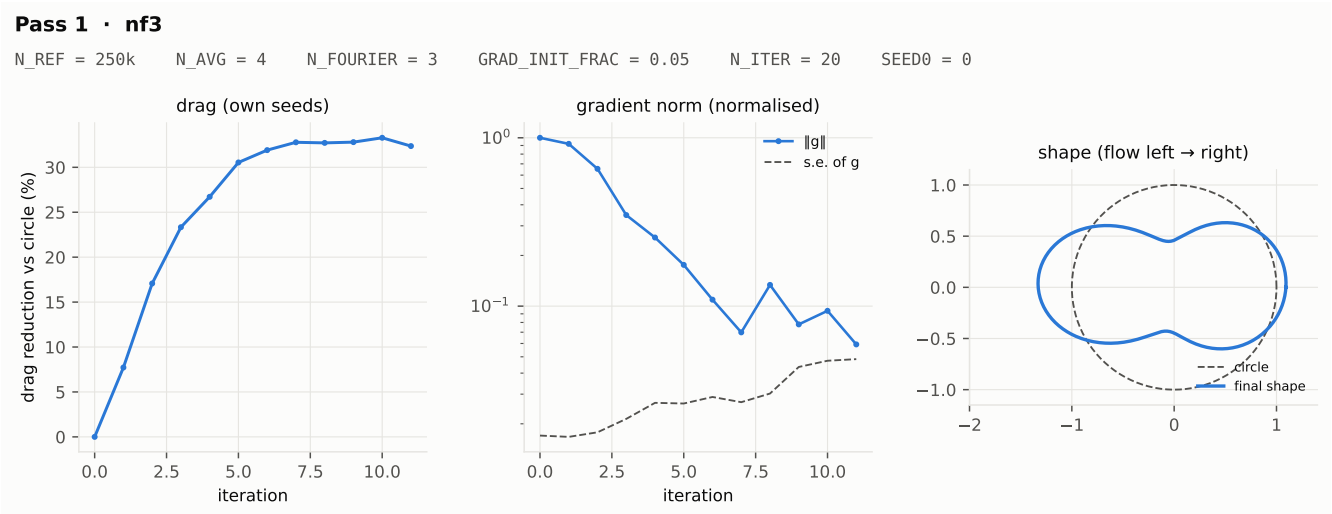


Figure 53: Pass 1, `nf3`. The run stopped at the noise limit after 11 iterations with 4 seeds. Its drag reduction was 32.4% relative to the circle on its own seeds and 32.6% relative to the circle on unseen seeds. It took 0.3 hours.

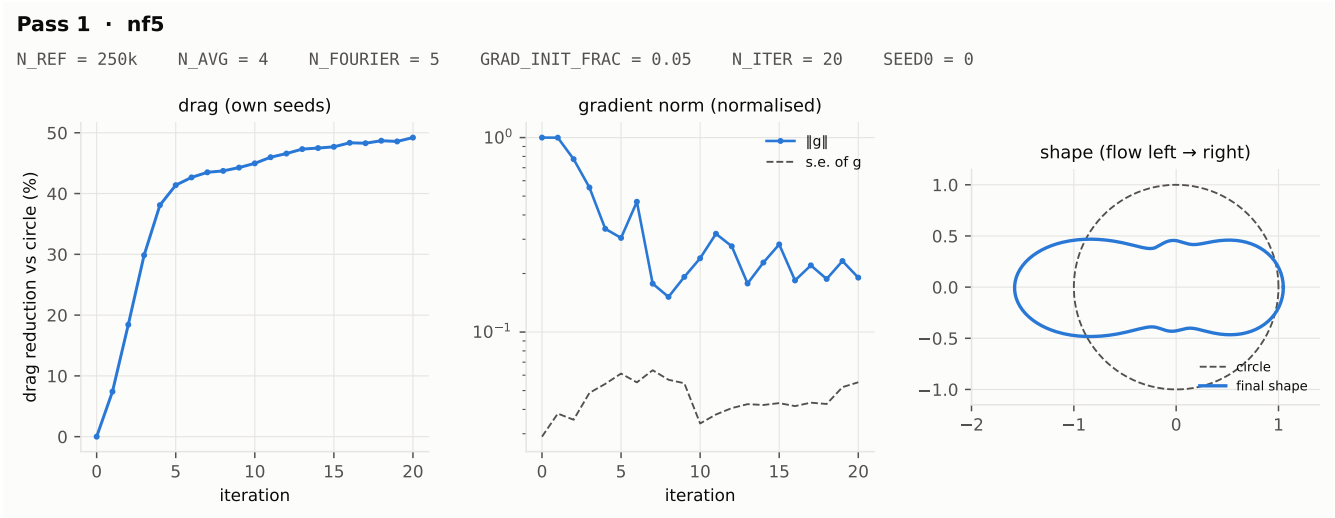


Figure 54: Pass 1, `nf5`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 49.2% relative to the circle on its own seeds and 49.0% relative to the circle on unseen seeds. It took 0.4 hours.

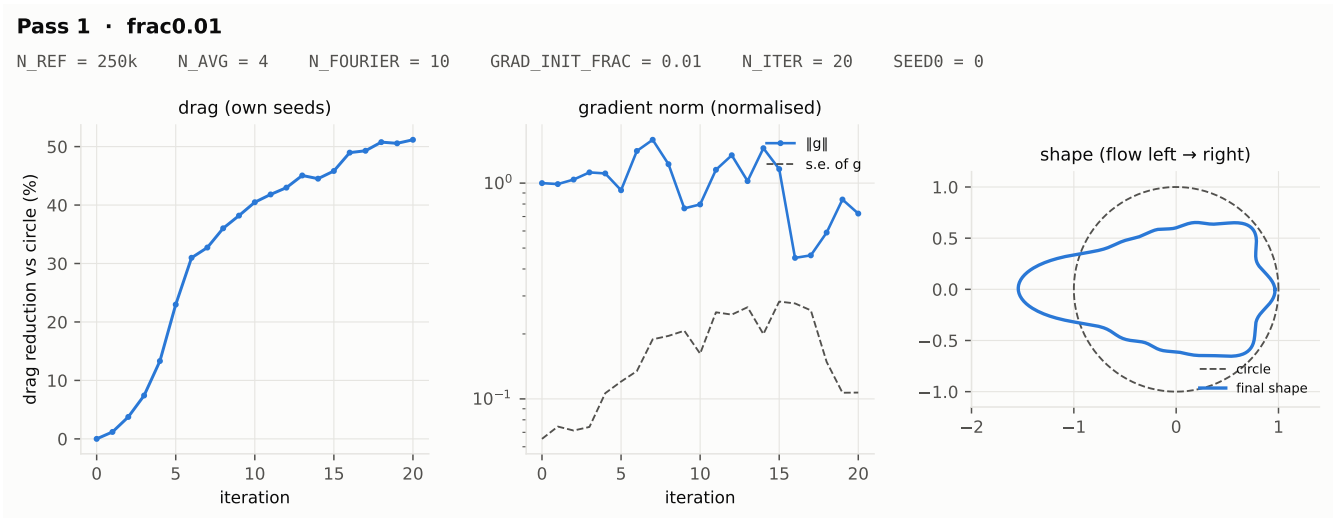


Figure 55: Pass 1, `frac0.01`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 51.2% relative to the circle on its own seeds and 51.5% relative to the circle on unseen seeds. It took 0.6 hours.

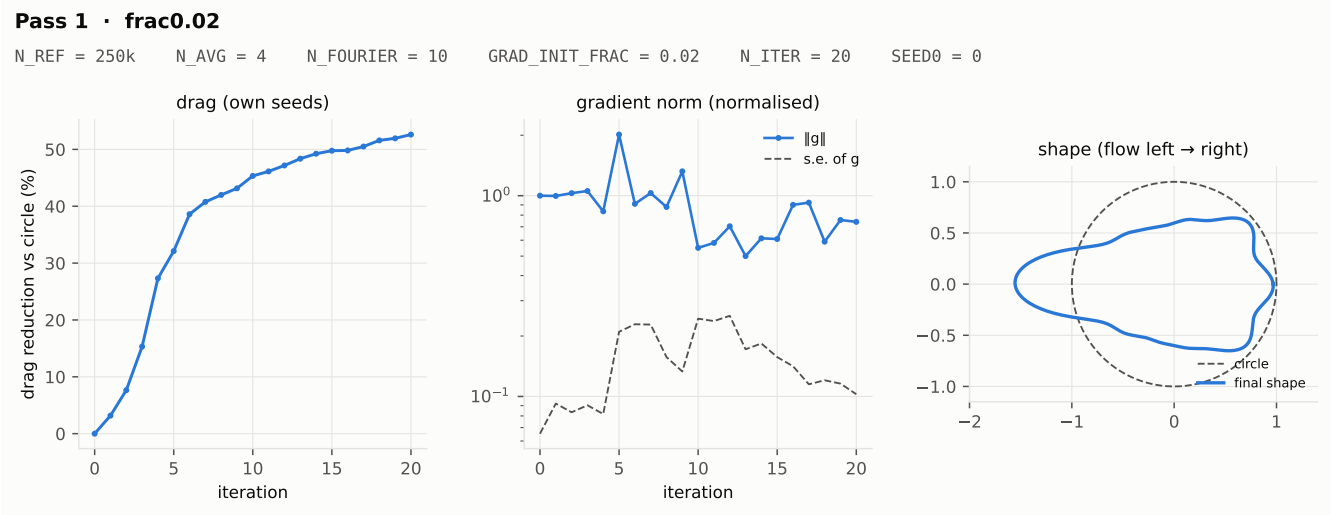


Figure 56: Pass 1, `frac0.02`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 52.6% relative to the circle on its own seeds and 52.4% relative to the circle on unseen seeds. It took 0.6 hours.

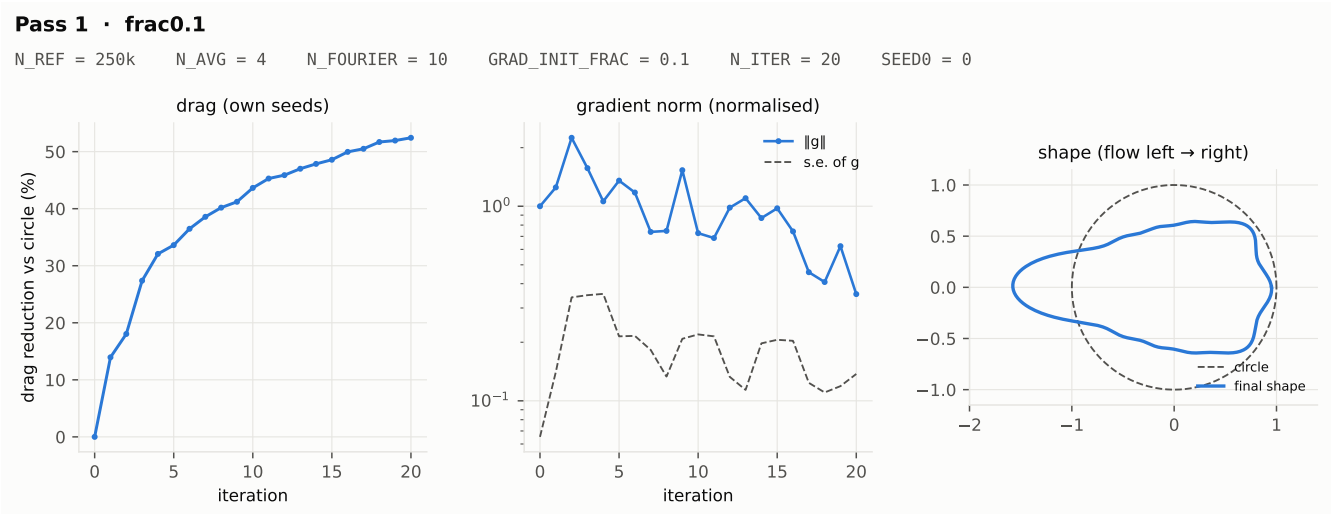


Figure 57: Pass 1, `frac0.1`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 52.4% relative to the circle on its own seeds and 52.6% relative to the circle on unseen seeds. It took 0.7 hours.

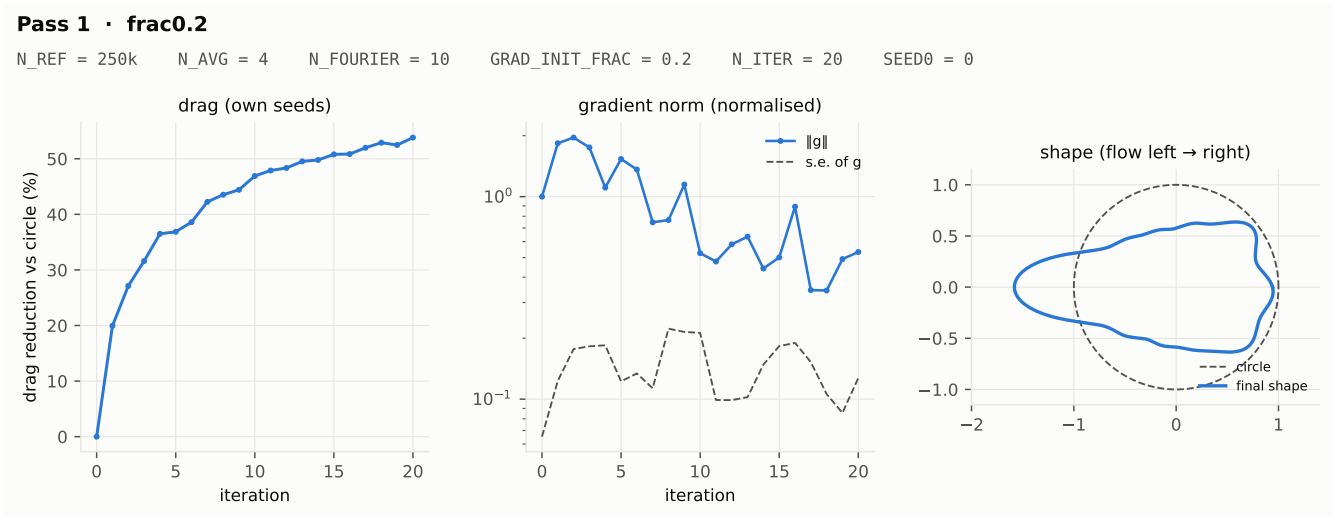


Figure 58: Pass 1, `frac0.2`. The run stopped at the iteration limit after 20 iterations with 4 seeds. Its drag reduction was 53.8% relative to the circle on its own seeds and 53.5% relative to the circle on unseen seeds. It took 0.6 hours.

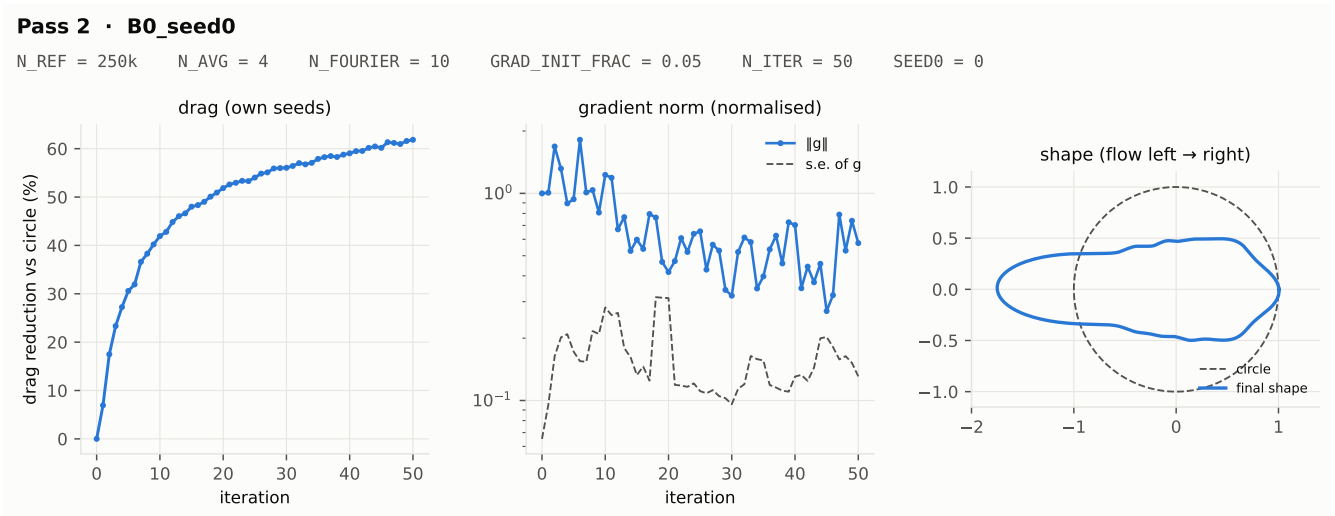


Figure 59: Pass 2, `B0_seed0`. The run stopped at the iteration limit after 50 iterations with 4 seeds. Its drag reduction was 61.8% relative to the circle on its own seeds and 62.0% relative to the circle on unseen seeds. It took 1.6 hours.

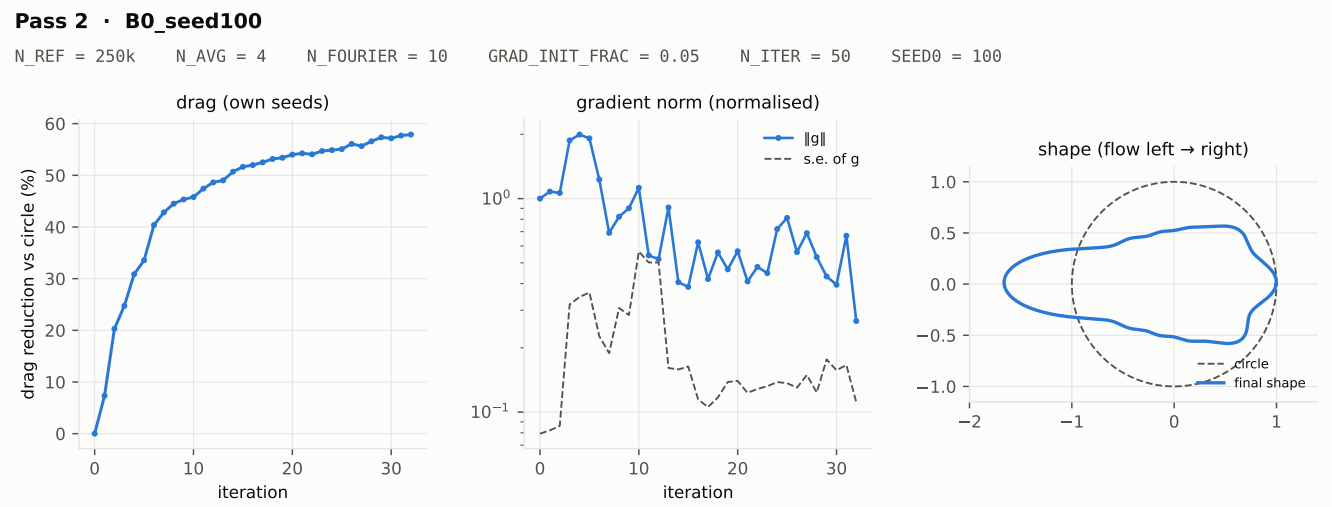


Figure 60: Pass 2, `B0_seed100`. The run stopped at the noise limit after 32 iterations with 4 seeds. Its drag reduction was 57.9% relative to the circle on its own seeds and 58.3% relative to the circle on unseen seeds. It took 1.1 hours.

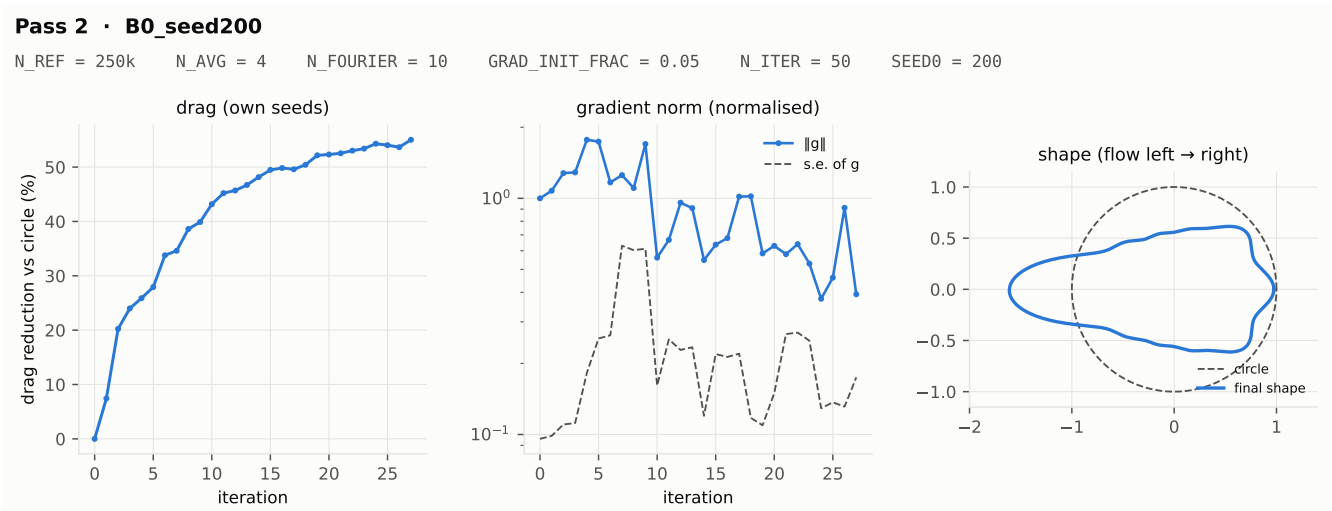


Figure 61: Pass 2, `B0_seed200`. The run stopped at the noise limit after 27 iterations with 4 seeds. Its drag reduction was 55.0% relative to the circle on its own seeds and 55.4% relative to the circle on unseen seeds. It took 0.9 hours.

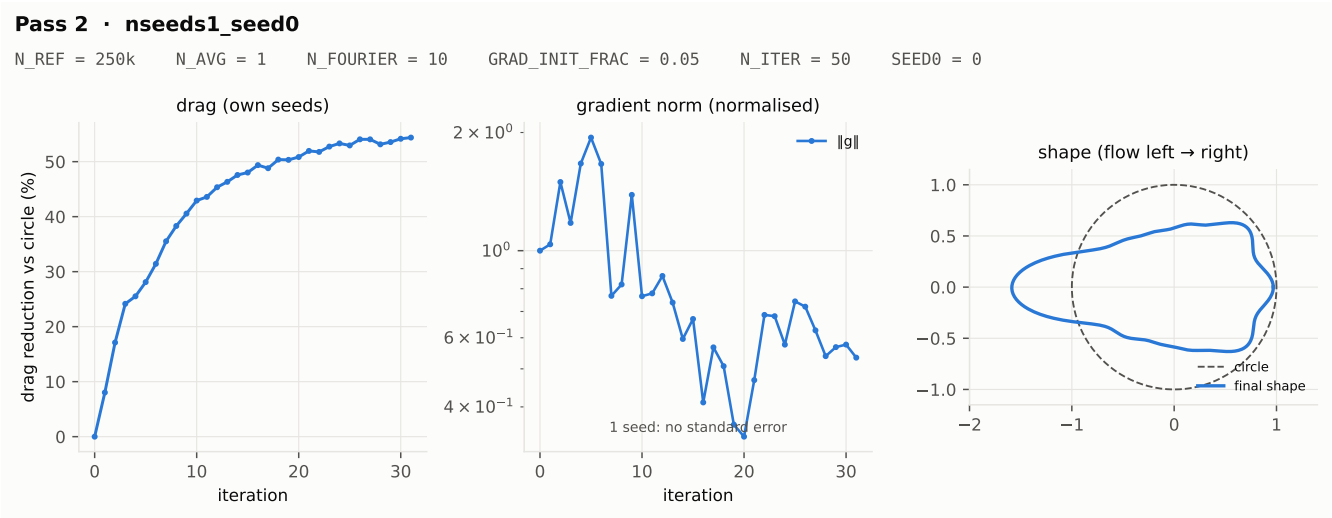


Figure 62: Pass 2, `nseeds1_seed0`. The run stopped at the noise limit after 31 iterations with 1 seeds. Its drag reduction was 54.4% relative to the circle on its own seeds and 53.7% relative to the circle on unseen seeds. It took 0.9 hours.

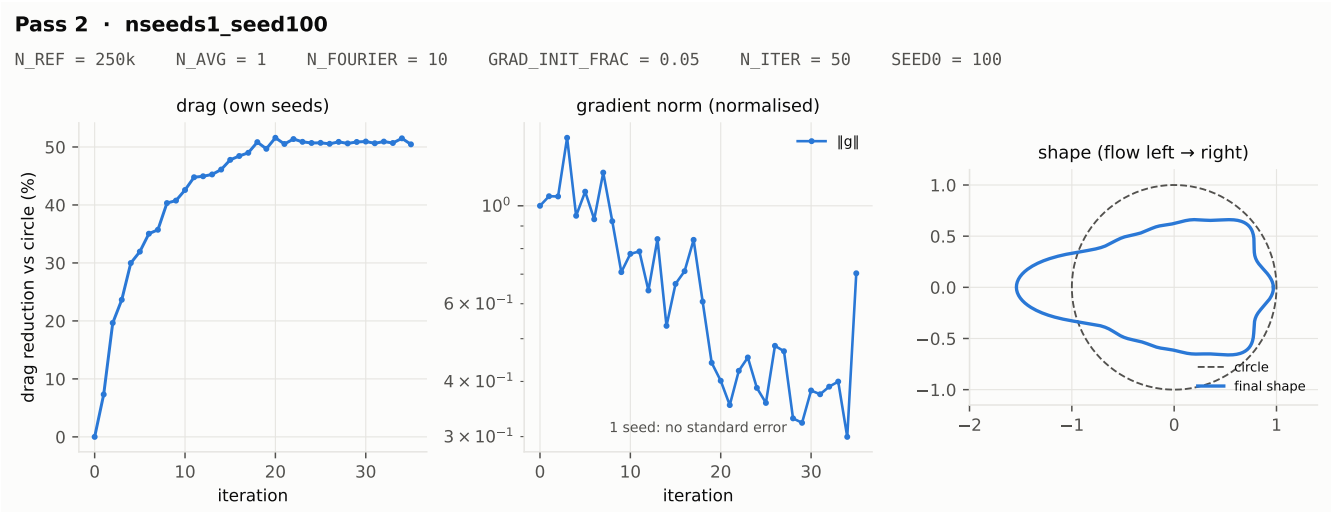


Figure 63: Pass 2, `nseeds1_seed100`. The run stopped at the noise limit after 35 iterations with 1 seeds. Its drag reduction was 50.5% relative to the circle on its own seeds and 51.2% relative to the circle on unseen seeds. It took 1.0 hours.

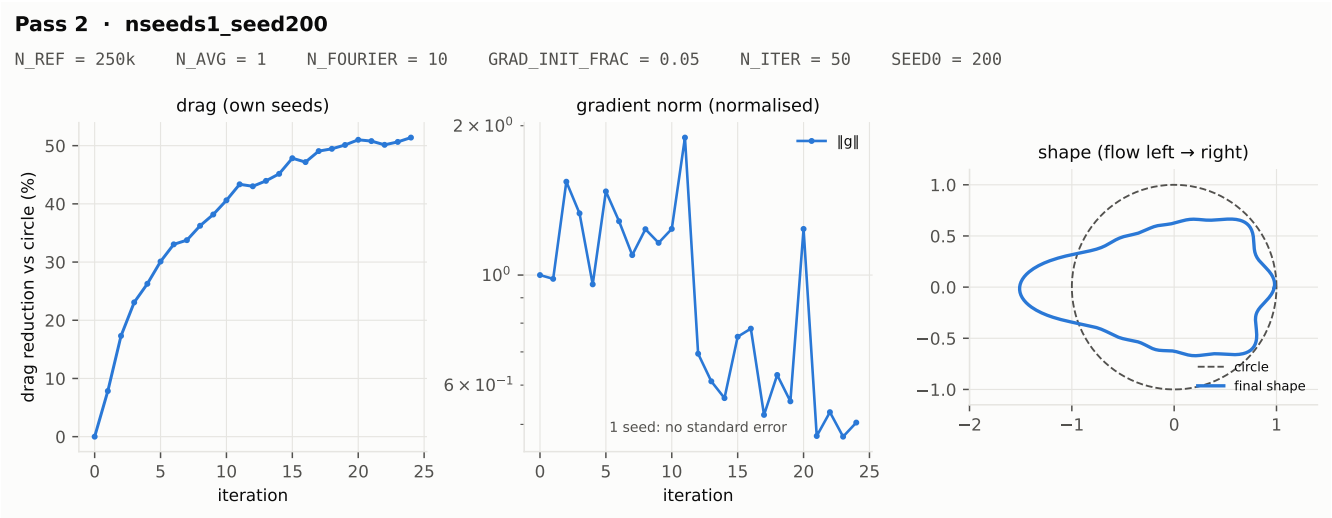


Figure 64: Pass 2, `nseeds1_seed200`. The run stopped at the noise limit after 24 iterations with 1 seeds. Its drag reduction was 51.4% relative to the circle on its own seeds and 50.4% relative to the circle on unseen seeds. It took 0.8 hours.

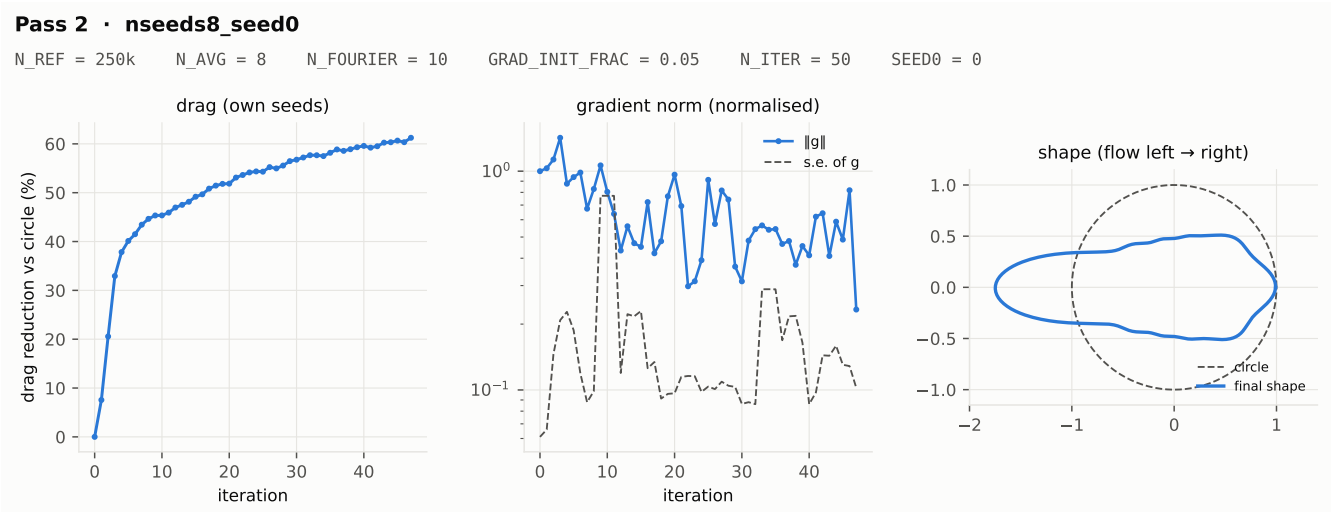


Figure 65: Pass 2, `nseeds8_seed0`. The run stopped at the noise limit after 47 iterations with 8 seeds. Its drag reduction was 61.2% relative to the circle on its own seeds and 61.4% relative to the circle on unseen seeds. It took 1.5 hours.

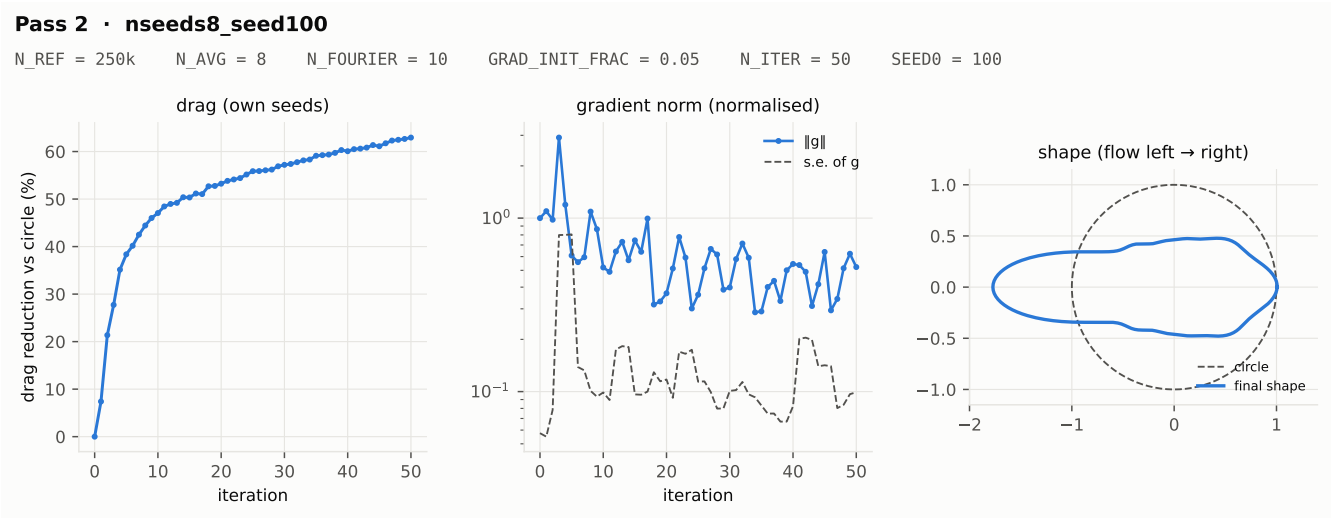


Figure 66: Pass 2, `nseeds8_seed100`. The run stopped at the iteration limit after 50 iterations with 8 seeds. Its drag reduction was 62.9% relative to the circle on its own seeds and 62.9% relative to the circle on unseen seeds. It took 1.5 hours.

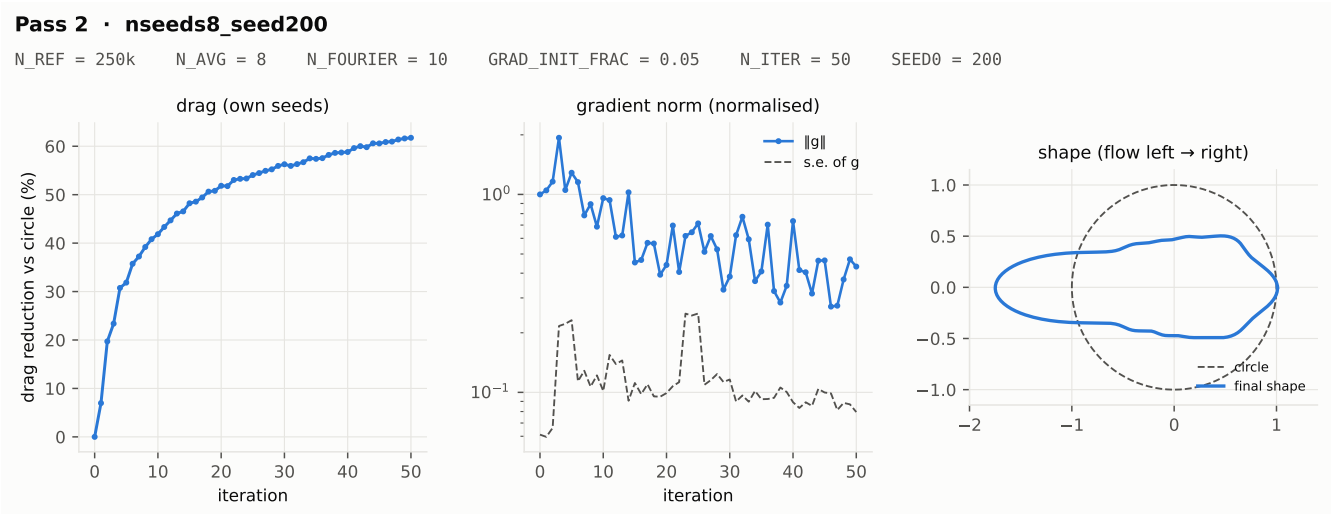


Figure 67: Pass 2, `nseeds8_seed200`. The run stopped at the iteration limit after 50 iterations with 8 seeds. Its drag reduction was 61.7% relative to the circle on its own seeds and 62.2% relative to the circle on unseen seeds. It took 1.7 hours.

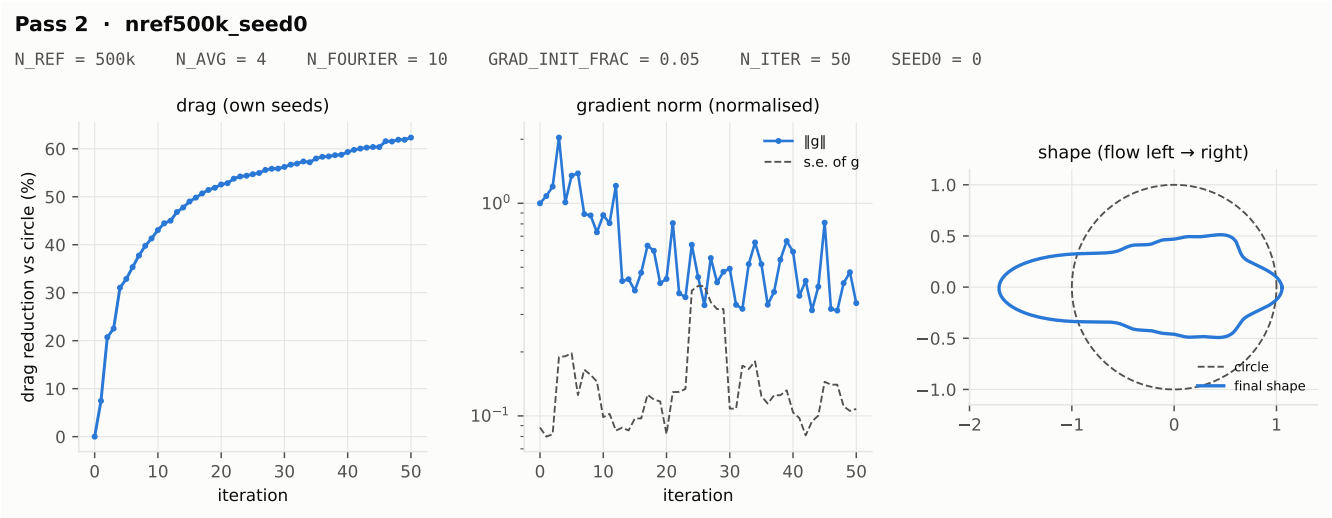


Figure 68: Pass 2, `nref500k_seed0`. The run stopped at the iteration limit after 50 iterations with 4 seeds. Its drag reduction was 62.3% relative to the circle on its own seeds and 62.7% relative to the circle on unseen seeds. It took 2.7 hours.

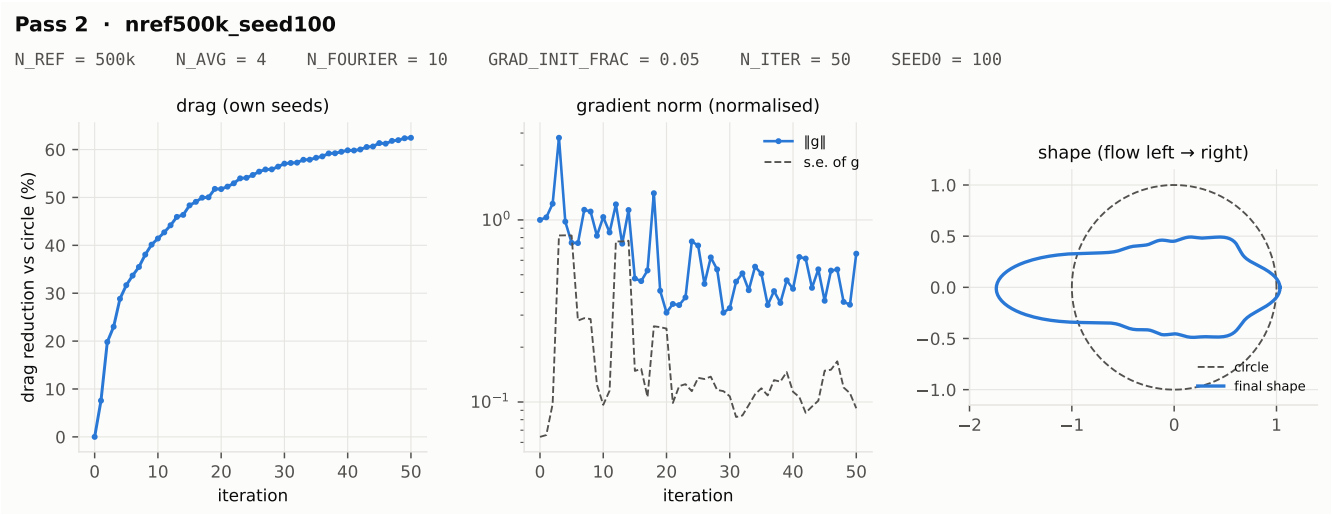


Figure 69: Pass 2, `nref500k_seed100`. The run stopped at the iteration limit after 50 iterations with 4 seeds. Its drag reduction was 62.5% relative to the circle on its own seeds and 62.7% relative to the circle on unseen seeds. It took 2.8 hours.

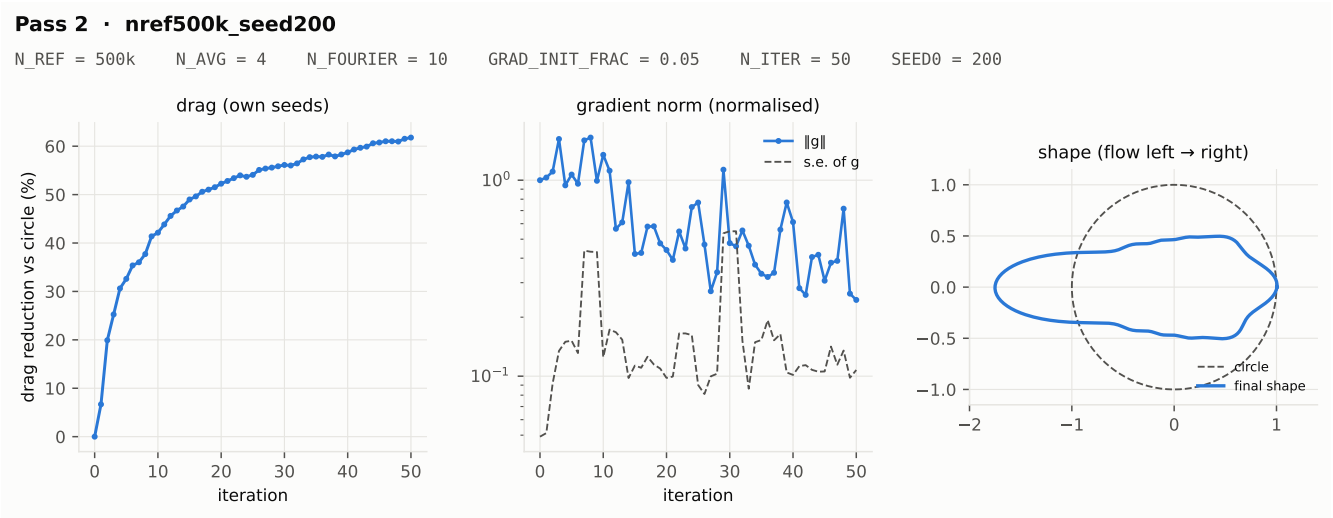


Figure 70: Pass 2, `nref500k_seed200`. The run stopped at the iteration limit after 50 iterations with 4 seeds. Its drag reduction was 61.8% relative to the circle on its own seeds and 62.3% relative to the circle on unseen seeds. It took 2.8 hours.

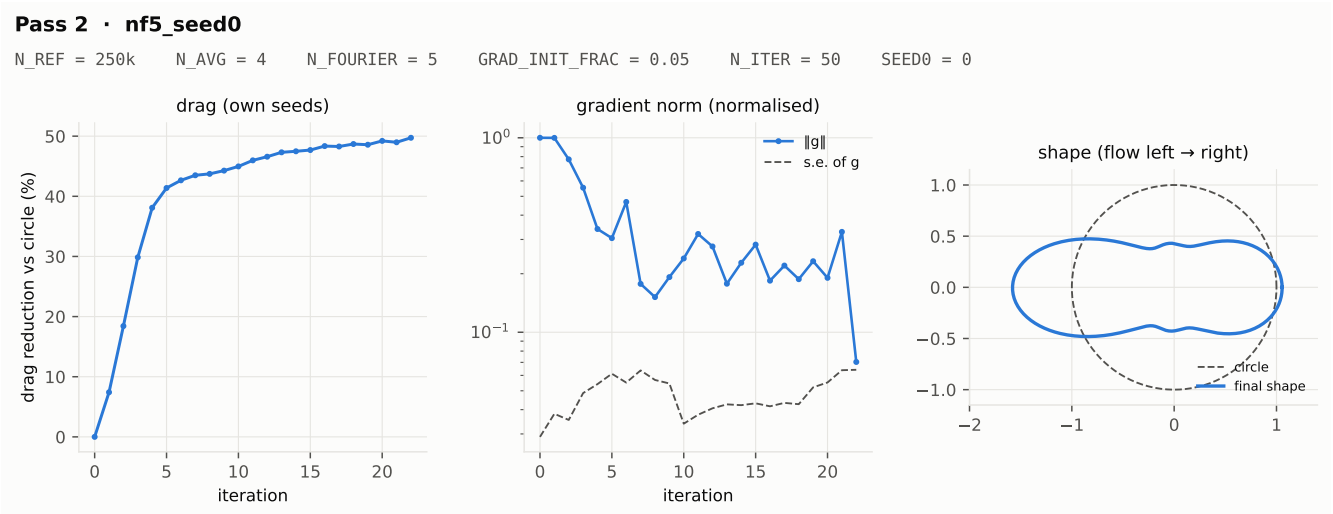


Figure 71: Pass 2, `nf5_seed0`. The run stopped at the noise limit after 22 iterations with 4 seeds. Its drag reduction was 49.7% relative to the circle on its own seeds and 49.4% relative to the circle on unseen seeds. It took 0.6 hours.

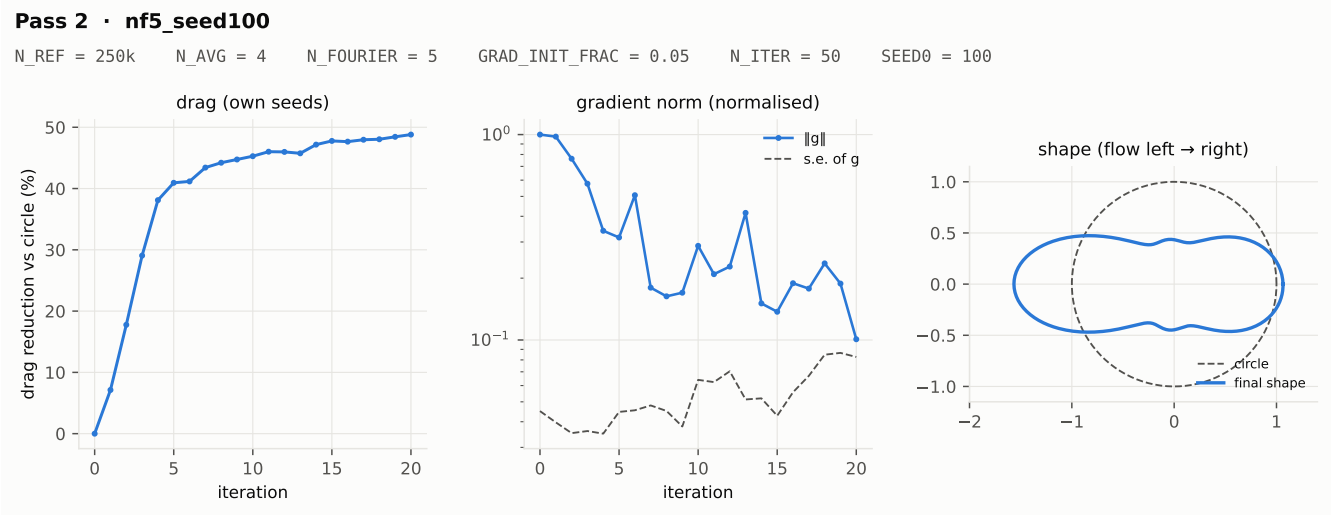


Figure 72: Pass 2, `nf5_seed100`. The run stopped at the noise limit after 20 iterations with 4 seeds. Its drag reduction was 48.8% relative to the circle on its own seeds and 49.5% relative to the circle on unseen seeds. It took 0.4 hours.

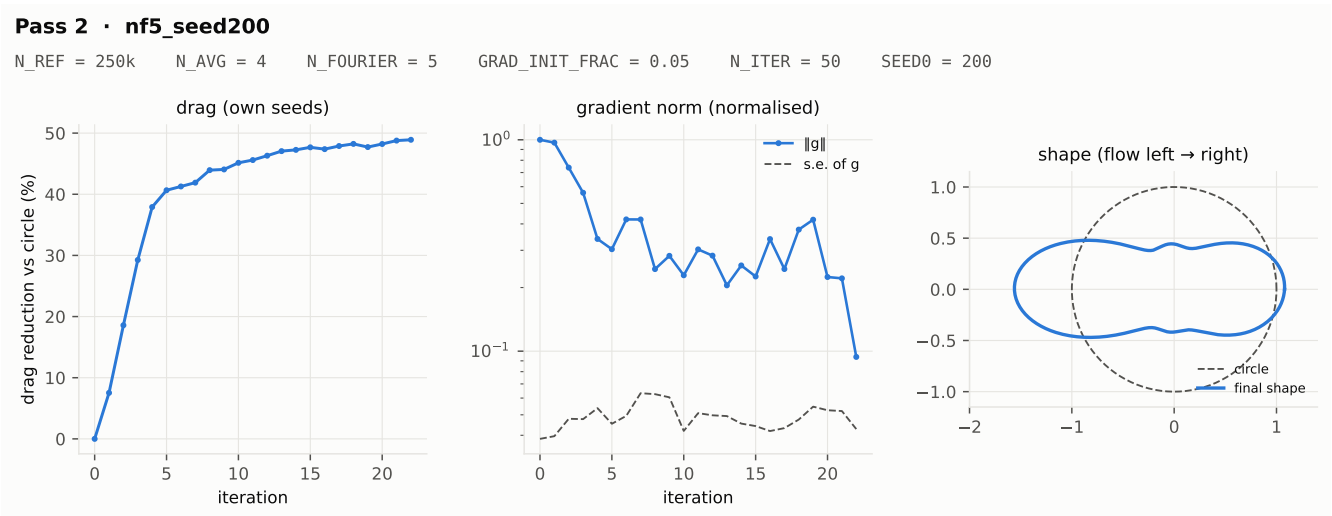


Figure 73: Pass 2, `nf5_seed200`. The run stopped at the noise limit after 22 iterations with 4 seeds. Its drag reduction was 48.9% relative to the circle on its own seeds and 49.6% relative to the circle on unseen seeds. It took 0.4 hours.

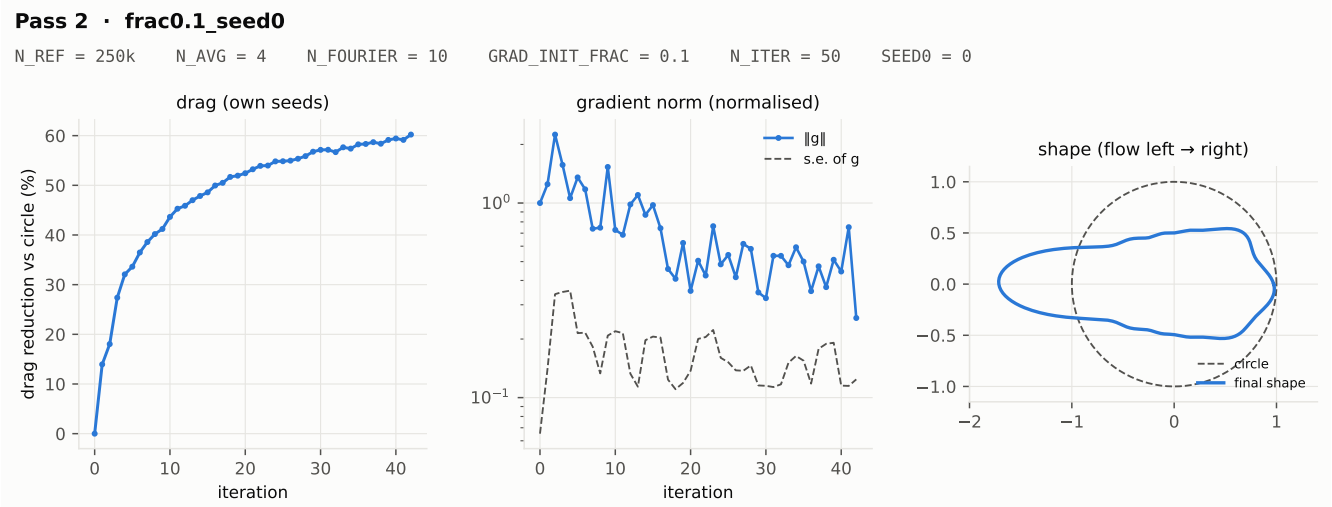


Figure 74: Pass 2, `frac0.1_seed0`. The run stopped at the noise limit after 42 iterations with 4 seeds. Its drag reduction was 60.2% relative to the circle on its own seeds and 60.1% relative to the circle on unseen seeds. It took 1.3 hours.

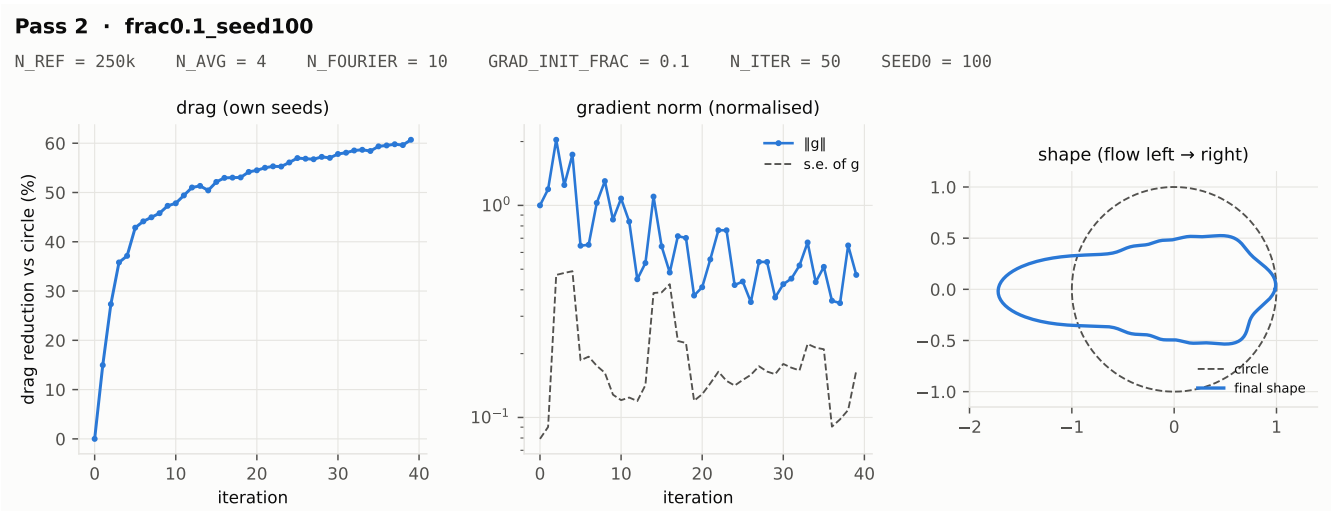


Figure 75: Pass 2, `frac0.1_seed100`. The run stopped at the noise limit after 39 iterations with 4 seeds. Its drag reduction was 60.7% relative to the circle on its own seeds and 60.4% relative to the circle on unseen seeds. It took 1.2 hours.

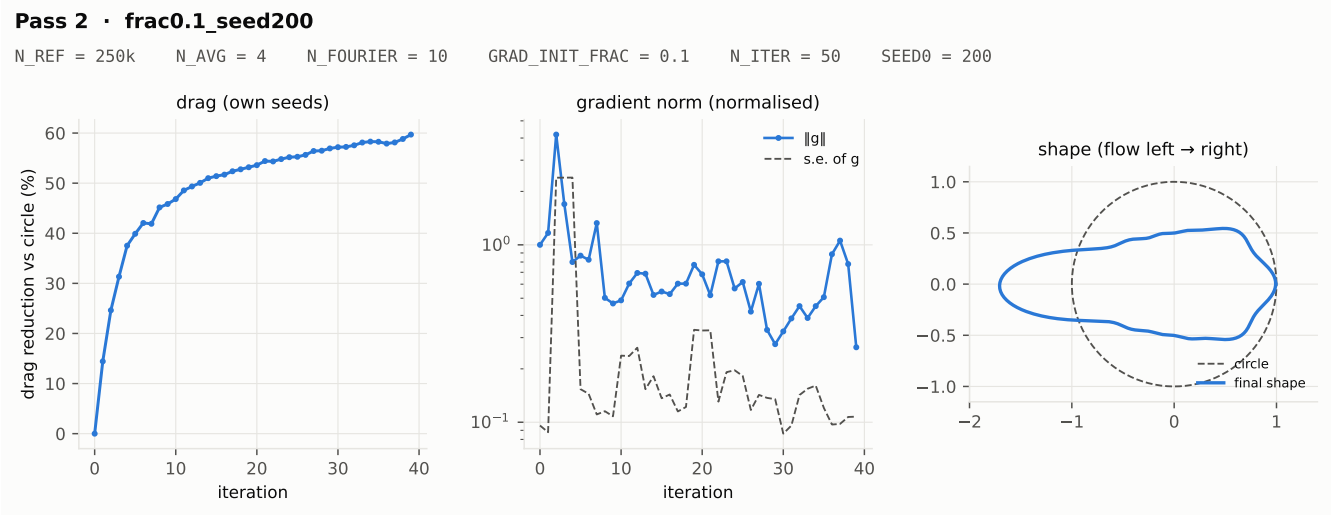


Figure 76: Pass 2, `frac0.1_seed200`. The run stopped at the noise limit after 39 iterations with 4 seeds. Its drag reduction was 59.7% relative to the circle on its own seeds and 59.6% relative to the circle on unseen seeds. It took 1.2 hours.

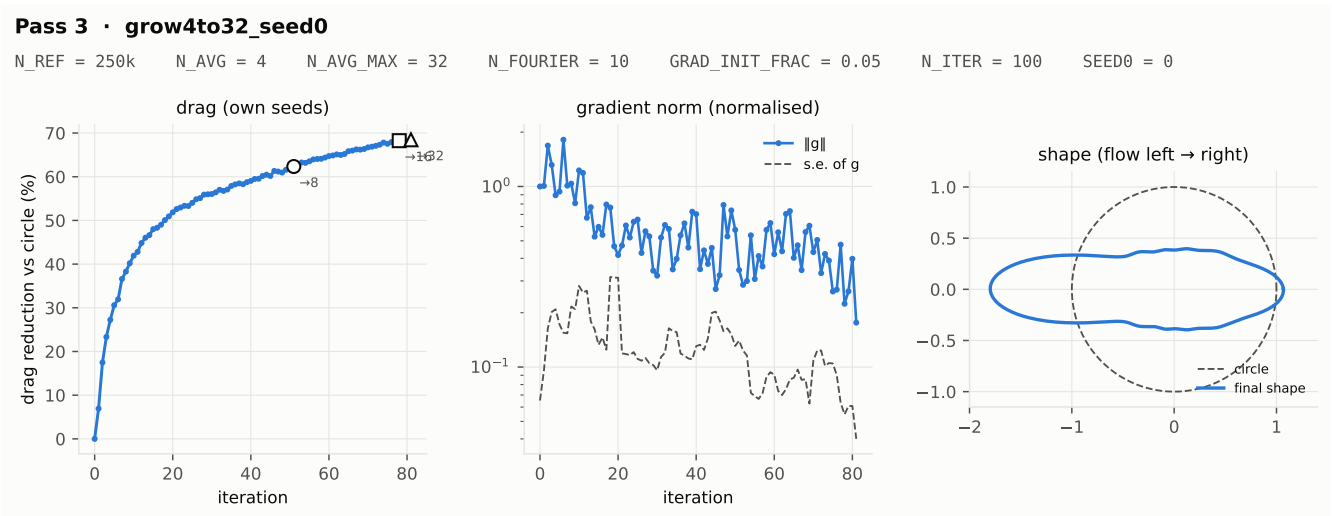


Figure 77: Pass 3, `grow4to32_seed0`. The run stopped at the noise limit after 81 iterations with 32 seeds. Its drag reduction was 68.5% relative to the circle on its own seeds and 68.1% relative to the circle on unseen seeds. It took 4.1 hours.

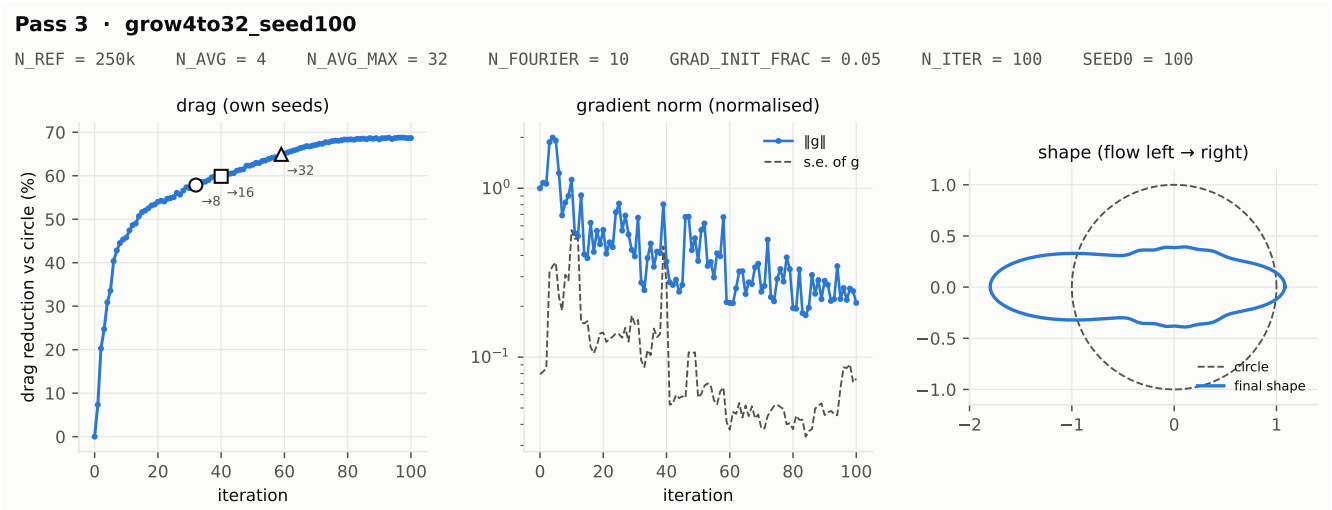


Figure 78: Pass 3, `grow4to32_seed100`. The run stopped at the iteration limit after 100 iterations with 32 seeds. Its drag reduction was 68.7% relative to the circle on its own seeds and 69.0% relative to the circle on unseen seeds. It took 11.6 hours.

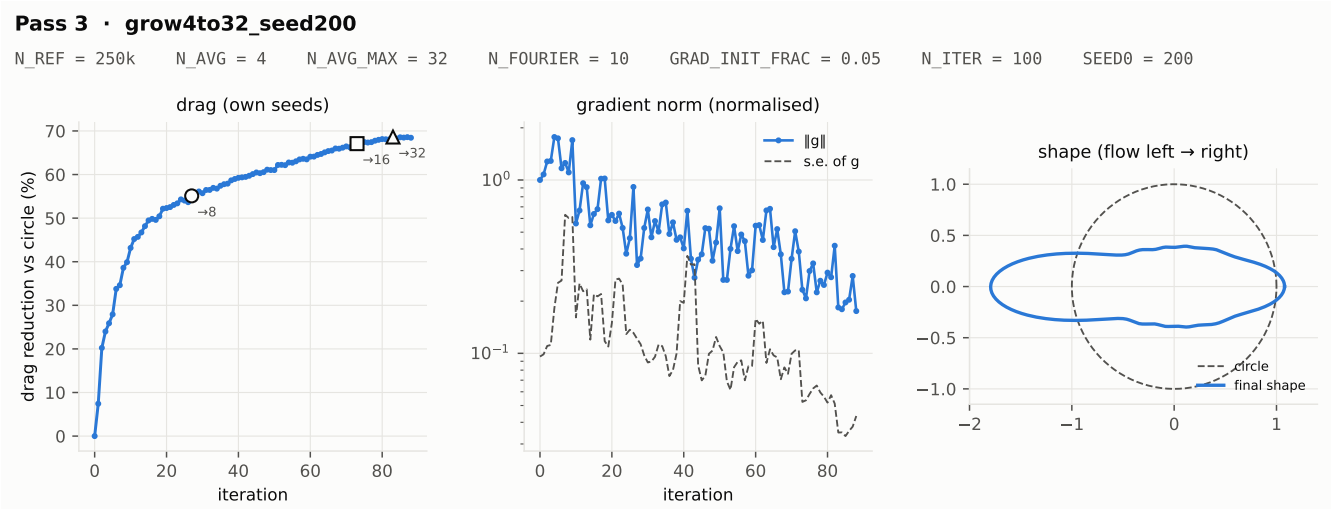


Figure 79: Pass 3, `grow4to32_seed200`. The run stopped at the noise limit after 88 iterations with 32 seeds. Its drag reduction was 68.4% relative to the circle on its own seeds and 68.6% relative to the circle on unseen seeds. It took 6.1 hours.

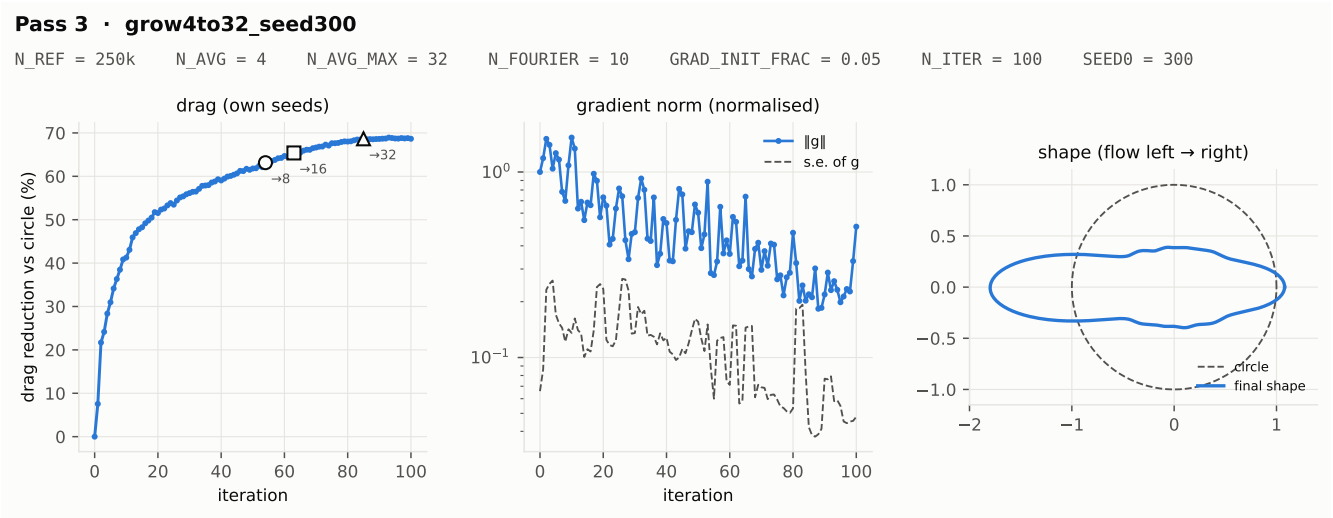


Figure 80: Pass 3, `grow4to32_seed300`. The run stopped at the iteration limit after 100 iterations with 32 seeds. Its drag reduction was 68.7% relative to the circle on its own seeds and 68.5% relative to the circle on unseen seeds. It took 8.3 hours.

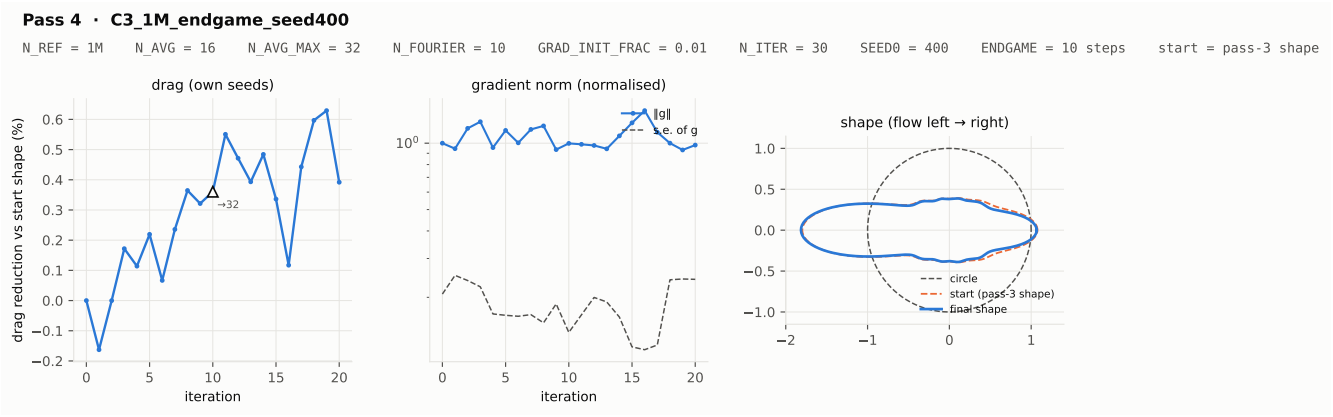


Figure 81: Pass 4, `C3_1M_endgame_seed400`. The run stopped at the noise limit after 20 iterations with 32 seeds. Its drag reduction was 0.4% relative to the start shape on its own seeds and 68.9% relative to the circle on unseen seeds. It took 6.3 hours.